Mathematical Models in Biotechnology

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Abstract

Mathematical models in biotechnology span a spectrum from fully mechanistic ordinary-differential-equation (ODE) systems grounded in conservation laws and reaction kinetics to purely data-driven deep sequence architectures that infer temporal structure from observations alone. This chapter surveys that spectrum with a worked example: a CHO fed-batch monoclonal-antibody bioreactor revisited through four complementary lenses. We begin with kinetic ODE models—Michaelis–Menten enzyme kinetics, Monod growth, and Luedeking–Piret product formation—integrated into a seven-state fed-batch simulator producing the ground-truth trajectories used throughout. We then develop constraint-based flux balance analysis (FBA), illustrated on a urea-cycle model of HL-60 cells in which a linear program maximizes urea export subject to stoichiometric and capacity constraints. Models of gene expression are treated next: Hill-type transcriptional input functions, network motifs (negative autoregulation, coherent and incoherent feed-forward loops, Goodwin oscillator), and cybernetic resource-allocation models of diauxic growth. The deep-learning section derives Elman RNN, LSTM, and structured state-space (S4/HiPPO-LegS) architectures and demonstrates a head-to-head comparison on the CHO system in which S4 achieves lower test error on all seven process states. Finally, a hybrid residual-learning framework—a mechanistic ODE prior augmented by a learned discrepancy model—corrects parameter-induced model drift with substantially greater data efficiency than pure-mechanistic or pure-data-driven forecasters. Together, the examples argue that the productive frontier of biotechnology lies in mechanism-informed machine learning: hybrid architectures that preserve interpretability and extrapolation capability of first-principles models while capturing the flexibility of learned components.

1 Introduction

Mathematical models are indispensable tools in modern biotechnology. At their most general, they are formal statements of what we believe about a biological or biochemical system: how metabolites accumulate, how genes are regulated, how product titer evolves over the course of a fed-batch culture. The purpose of a model shapes its construction in fundamental ways. A model built for *prediction* must generalize beyond the conditions used to fit it; one built for *design* must expose which process variables, when manipulated, move a performance objective in the desired direction; a model used for *control* must be computable fast enough to act within the timescales of the process it governs; and a model aimed at *mechanistic understanding* must be interpretable in terms of biology. These purposes are not mutually exclusive, but the tension among them drives much of the methodological diversity described in this chapter.

A first organizing principle is the degree to which model parameters carry physical meaning. In a *mechanistic* (or first-principles) model, each parameter corresponds to a measurable quantity: an enzymatic rate constant, a stoichiometric coefficient, a binding affinity. The governing equations are derived from conservation laws, reaction kinetics, or documented molecular mechanisms.

Such models can, in principle, extrapolate beyond the training envelope because the underlying physics continues to hold; their weakness is that constructing and fitting them requires substantial biological prior knowledge and high-quality experimental data. At the other end of the spectrum, *data-driven* models learn structure directly from observations, making few assumptions about mechanism. Deep neural networks and recurrent architectures belong to this class: they can represent highly nonlinear, high-dimensional dynamics with minimal domain knowledge, but their parameters are not individually interpretable, and their ability to extrapolate outside the distribution of training data is not guaranteed. Between these extremes lies a productive *hybrid* territory in which mechanistic structure constrains or initializes a learned component, combining the interpretability and extrapolation properties of the first-principles approach with the representational power of the data-driven one. This chapter is organized to reflect that spectrum, moving from purely mechanistic formulations toward learned and hybrid models.

Four themes recur throughout and deserve explicit introduction here. *Identifiability* refers to whether the parameters of a model can in principle be uniquely recovered from the available data. Structural identifiability is a property of the model equations alone; practical (or numerical) identifiability depends additionally on the quantity and quality of observations. A model that is not identifiable cannot be reliably calibrated, and predictions drawn from it carry unquantified uncertainty. *Parameter estimation* is the computational process of fitting model parameters to experimental data, typically by minimizing a weighted residual or maximizing a likelihood function; it is distinct from model selection, which chooses among candidate structures. *Calibration versus validation* distinguishes the data used to fit a model from the data used to assess whether it generalizes: a model that describes its calibration data well may still fail on independent validation experiments, particularly when the two experiment sets differ in operating conditions or scale. Finally, *extrapolation* is the central reason mechanistic structure continues to matter even as data-driven methods mature. A model that has internalized a conservation law or a reaction mechanism retains predictive power outside the conditions it was trained on; a model that has merely interpolated a training manifold does not.

Two experimental systems appear repeatedly as concrete illustrations. The first is a *CHO fed-batch monoclonal-antibody (mAb) bioreactor*. Chinese hamster ovary (CHO) cells are the dominant production platform for therapeutic antibodies, and the fed-batch operation mode is standard at commercial scale. This system is introduced in Section 2 as a structured kinetic ODE model in which cell growth, substrate consumption, and product formation are linked through measurable rate expressions. The same bioreactor appears in Sections 5 and 6 as the target of data-driven time-series forecasting and hybrid grey-box modeling, making it possible to compare mechanistic and learned representations on identical biological ground. The second recurring system is a set of *E. coli gene-regulatory circuits*. Prokaryotic gene networks are well characterized, small enough to analyze analytically, and rich enough in qualitative behavior (oscillation, adaptation) to illustrate fundamental concepts in network dynamics. These circuits serve as the substrate for the network-motif analysis in Section 4, which also develops cybernetic models of regulatory resource allocation applied to diauxic growth on mixed substrates.

The remainder of this chapter is structured as follows. Section 2 introduces kinetic and mechanistic ordinary-differential-equation models, covering mass-action and Michaelis–Menten kinetics, structured cell-growth models, and the CHO fed-batch example. Section 3 presents flux balance analysis, the constraint-based stoichiometric framework that enables genome-scale metabolic modeling without requiring kinetic rate expressions. Section 4 treats models of gene expression, with emphasis on *E. coli* network motifs and transcription-factor dynamics, and cybernetic modeling of regulatory resource allocation during diauxic growth on mixed substrates. Section 5 surveys deep time-series architectures—recurrent neural networks, long short-term memory networks, and structured state-space models—applied to bioprocess forecasting. Section 6 examines hybrid modeling strategies that embed mechanistic knowledge within learned representations, and closes with an outlook on open problems and emerging directions.

2 Kinetic and Mechanistic Models

Mathematical descriptions of biochemical reaction rates underpin virtually every quantitative model in biotechnology, from single-enzyme assays to whole-culture fed-batch reactors. This section develops the principal rate-law frameworks in order of increasing complexity: single-enzyme Michaelis–Menten kinetics, Monod growth kinetics for microbial and mammalian cell cultures, mass-action descriptions appropriate for signaling and coagulation networks, the Luedeking–Piret relationship for product formation, and finally a seven-state fed-batch model for monoclonal antibody (mAb) production in Chinese hamster ovary (CHO) cells that integrates all of these elements.

2.1 Enzyme Kinetics: the Michaelis–Menten Rate Law

Consider a single enzyme E that binds substrate S to form an enzyme–substrate complex ES, which either dissociates back to $E + S$ or proceeds irreversibly to release product P and regenerate free enzyme:

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P. \tag{1}$$

Under the quasi-steady-state assumption (QSSA)—that ES reaches a pseudo-steady state on a time scale much shorter than the depletion of S—the reaction velocity is (see Appendix, Section A.1 for the full derivation):

$$v = \frac{V_{\max}[S]}{K_M + [S]}, \tag{2}$$

where $V_{\max} = k_2[E]_T$ is the limiting velocity at saturating substrate and $K_M = (k_{-1} + k_2)/k_1$ is the Michaelis constant [1]. The parameter K_M equals the substrate concentration at which the rate is half-maximal; it therefore quantifies the apparent affinity of the enzyme for its substrate. At low substrate concentrations ($[S] \ll K_M$) the rate is first-order in $[S]$, while at high concentrations ($[S] \gg K_M$) it saturates at $V_{\max}$. This hyperbolic saturation behavior, arising from the finite pool of enzyme, is the hallmark of enzyme kinetics and recurs throughout the models developed in this chapter.

2.2 Monod Growth Kinetics

At the scale of a cell population the analogue of the Michaelis–Menten rate law is the Monod equation [2]. Monod observed empirically that the specific growth rate μ of a culture growing on a single limiting substrate S follows:

$$\mu = \mu_{\max} \frac{S}{K_S + S}, \tag{3}$$

where $\mu_{\max}$ is the maximum specific growth rate (h^{-1}) and K_S is the half-saturation constant (concentration units). The formal analogy with Eq. (2) is not coincidental: the saturation of the cellular growth machinery by nutrient transporters and metabolic enzymes, each individually obeying Michaelis–Menten kinetics, collectively produces the Monod hyperbolic form at the population scale.

When growth is co-limited by two substrates S_1 and S_2 the limitation terms are multiplied:

$$\mu = \mu_{\max} \frac{S_1}{K_{S_1} + S_1} \frac{S_2}{K_{S_2} + S_2}, \tag{4}$$

expressing the assumption that limitation by each nutrient acts independently and multiplicatively. Inhibitory byproducts reduce growth through additional inhibition factors of the form $K_I/(K_I + I)$, where I is the inhibitor concentration and K_I is the inhibition constant; high K_I corresponds to weak inhibition.

2.3 Mass-Action Kinetics

When individual molecular species and their stoichiometric interactions must be resolved explicitly, mass-action kinetics provide the fundamental framework. For an elementary reaction $A+B \to C$ the rate is $r = k[A][B]$, with the rate constant k encoding the probability of a productive collision per unit time per unit concentration. Systems of mass-action ODEs arise naturally from biochemical reaction networks: each reaction contributes a term proportional to the product of its reactant concentrations.

Mass-action models are indispensable when the kinetic order or mechanism of each step is known and the network is small enough to be parameterized. The Michaelis–Menten mechanism (Eq. 1) is itself a mass-action system at the level of the individual steps k_1, k_{-1}, and k_2; the QSSA collapses it to the simpler lumped form in Eq. (2). For large networks comprising tens to hundreds of species, however, parameter estimation becomes the central challenge, and reduced or hierarchical representations such as Monod or Luedeking–Piret kinetics are preferred.

2.4 Luedeking–Piret Product Formation

The specific rate of product formation in a fermentation often has two contributions: one coupled to growth and one that persists in the absence of growth [3]. The Luedeking–Piret relationship captures both:

$$q_P = \alpha\, \mu + \beta, \tag{5}$$

where q_P is the specific productivity (mass of product per unit biomass per unit time), α is the growth-associated coefficient, and β is the non-growth-associated coefficient. Primary metabolites such as ethanol are predominantly growth-associated ($\alpha \gg \beta$), while secondary metabolites and many therapeutic proteins exhibit significant non-growth contributions. The population-level product accumulation rate is $q_P X$, where X is the biomass concentration.

2.5 CHO Fed-Batch mAb Production: a Worked Example

To illustrate how enzyme-level kinetics, population-scale Monod growth, and Luedeking–Piret product formation combine in an industrially relevant setting, we present a seven-state ODE model for mAb production in a CHO fed-batch culture. The state vector is

$$\mathbf{u} = [V,\ X,\ \text{Glc},\ \text{Gln},\ \text{Lac},\ \text{Amm},\ \text{mAb}]^\top, \tag{6}$$

with units L, g_{DW}/L, mM, mM, mM, mM, and mg/L, respectively.

Growth and Death Rates

The specific growth rate combines dual Monod limitation by glucose (Glc) and glutamine (Gln) with multiplicative inhibition by lactate (Lac) and ammonia (Amm):

$$\mu = \mu_{\max} \frac{\mathrm{Glc}}{K_{\mathrm{glc}} + \mathrm{Glc}} \frac{\mathrm{Gln}}{K_{\mathrm{gln}} + \mathrm{Gln}} \frac{K_{I,\mathrm{lac}}}{K_{I,\mathrm{lac}} + \mathrm{Lac}} \frac{K_{I,\mathrm{amm}}}{K_{I,\mathrm{amm}} + \mathrm{Amm}}. \tag{7}$$

The specific death rate μ_d is driven by the same byproducts, capturing the toxicity of lactate and ammonia accumulation:

$$\mu_d = k_d \frac{\mathrm{Lac}}{K_{D,\mathrm{lac}} + \mathrm{Lac}} \frac{\mathrm{Amm}}{K_{D,\mathrm{amm}} + \mathrm{Amm}}. \tag{8}$$

Specific Substrate Uptake and Byproduct Formation Rates

Yield coefficients relate substrate consumption and byproduct formation to growth:

$$q_{\mathrm{glc}} = \frac{\mu}{Y_{X/\mathrm{glc}}}, \quad q_{\mathrm{gln}} = \frac{\mu}{Y_{X/\mathrm{gln}}}, \quad q_{\mathrm{lac}} = Y_{\mathrm{lac/glc}}\, q_{\mathrm{glc}}, \quad q_{\mathrm{amm}} = Y_{\mathrm{amm/gln}}\, q_{\mathrm{gln}}, \tag{9}$$

where $Y_{X/\mathrm{glc}}$ and $Y_{X/\mathrm{gln}}$ are the biomass yields on glucose and glutamine ($\mathrm{g_{DW}}$/mmol), and $Y_{\mathrm{lac/glc}}$ and $Y_{\mathrm{amm/gln}}$ are the stoichiometric byproduct yields (mmol/mmol).

mAb Specific Productivity

The specific mAb productivity follows the Luedeking–Piret form (Eq. 5):

$$q_P = \alpha_P\, \mu + \beta_P, \tag{10}$$

with α_P in mg/$\mathrm{g_{DW}}$ and β_P in mg/$\mathrm{g_{DW}}$/h. The non-growth-associated term β_P captures the continued secretion of antibody by cells that have ceased dividing—an important contribution during the stationary phase of fed-batch operation.

Mass Balances

Let F denote the volumetric feed rate (L/h) and $D = F/V$ the resulting dilution rate. The feed supplies concentrated glucose and glutamine at concentrations $S_{\mathrm{glc},f}$ and $S_{\mathrm{gln},f}$. The governing ODEs are:

$$\frac{dV}{dt} = F, \tag{11a}$$

$$\frac{dX}{dt} = (\mu - \mu_d - D)\, X, \tag{11b}$$

$$\frac{d\,\mathrm{Glc}}{dt} = D\,(S_{\mathrm{glc},f} - \mathrm{Glc}) - q_{\mathrm{glc}}\, X, \tag{11c}$$

$$\frac{d\,\mathrm{Gln}}{dt} = D\,(S_{\mathrm{gln},f} - \mathrm{Gln}) - q_{\mathrm{gln}}\, X, \tag{11d}$$

$$\frac{d\,\mathrm{Lac}}{dt} = q_{\mathrm{lac}}\, X - D\,\mathrm{Lac}, \tag{11e}$$

$$\frac{d\,\mathrm{Amm}}{dt} = q_{\mathrm{amm}}\, X - D\,\mathrm{Amm}, \tag{11f}$$

$$\frac{d\,\mathrm{mAb}}{dt} = q_P\, X - D\,\mathrm{mAb}. \tag{11g}$$

Each balance has the standard fed-batch structure: a dilution term $D(\cdot)$ due to volume increase, plus a generation or consumption term proportional to biomass X.

Glucose-Triggered Feed Policy

The feed rate F follows a hysteretic on/off policy triggered by the glucose concentration. When Glc falls below a lower threshold $\mathrm{Glc}_{\min}$, the feed is turned on at rate $F_{\max}$; it is switched off once Glc rises above an upper threshold $\mathrm{Glc}_{\max}$:

$$F = \begin{cases} F_{\max} & \text{if Glc} \leq \mathrm{Glc}_{\min} \text{ (feed on)}, \\ 0 & \text{if Glc} \geq \mathrm{Glc}_{\max} \text{ (feed off)}. \end{cases} \tag{12}$$

This hysteretic switching maintains glucose within a target window, preventing both starvation and the Warburg-like overflow metabolism that elevates lactate at high glucose concentrations.

Simulated Dynamics

Figure 1 shows biomass and mAb trajectories simulated over a 240 h fed-batch run using the model equations above and literature parameter values (see text). At inoculation the initial glucose concentration is set below $\mathrm{Glc}_{\min}$, so the feed is already on at $t = 0$ and glucose rises during the first hours of the run before the hysteretic cycle begins. Thereafter the glucose-triggered hysteretic feed engages repeatedly—glucose declines as cells consume it, falls below $\mathrm{Glc}_{\min}$, the feed activates at $F_{\max}$ replenishing glucose to $\mathrm{Glc}_{\max}$, and then switches off, producing the characteristic sawtooth pattern and a corresponding step-wise increase in reactor volume V. Growth proceeds exponentially during the early phase while glucose and glutamine remain abundant. As byproducts accumulate, the inhibitory terms in Eq. (7) reduce μ and the death term μ_d rises, driving the culture toward a stationary phase. The non-growth-associated coefficient β_P in Eq. (10) sustains mAb secretion even after net growth ceases, so the product titer continues to rise through the final hours of the batch.

2.6 Sidebar: Large Mechanistic Reaction Networks—the Hockin–Mann Coagulation Cascade

Michaelis–Menten and mass-action rate laws scale naturally to networks of arbitrary complexity. A notable biomedical example is the Hockin–Mann model of the human blood coagulation cascade [4], which describes the extrinsic coagulation pathway through several dozen coupled ODEs and elementary reactions, each written in mass-action or Michaelis–Menten form. Parameters were estimated from purified-component biochemical data, and the model successfully reproduces the threshold behavior, amplification, and time course of thrombin generation observed in plasma assays. This example illustrates that mechanistic kinetic models are not limited to a handful of reactions: when the biochemistry is well characterized, models of comparable size can capture network-level behavior that is inaccessible to simpler empirical correlations.

3 Flux Balance Analysis

Kinetic models of the type developed in Section 2 require mechanistic rate expressions and numerical values for every kinetic parameter. For large metabolic networks – modern genome-scale reconstructions contain thousands of reactions and metabolites [5] – this level of mechanistic detail is rarely available. Flux balance analysis (FBA) circumvents this limitation by replacing differential

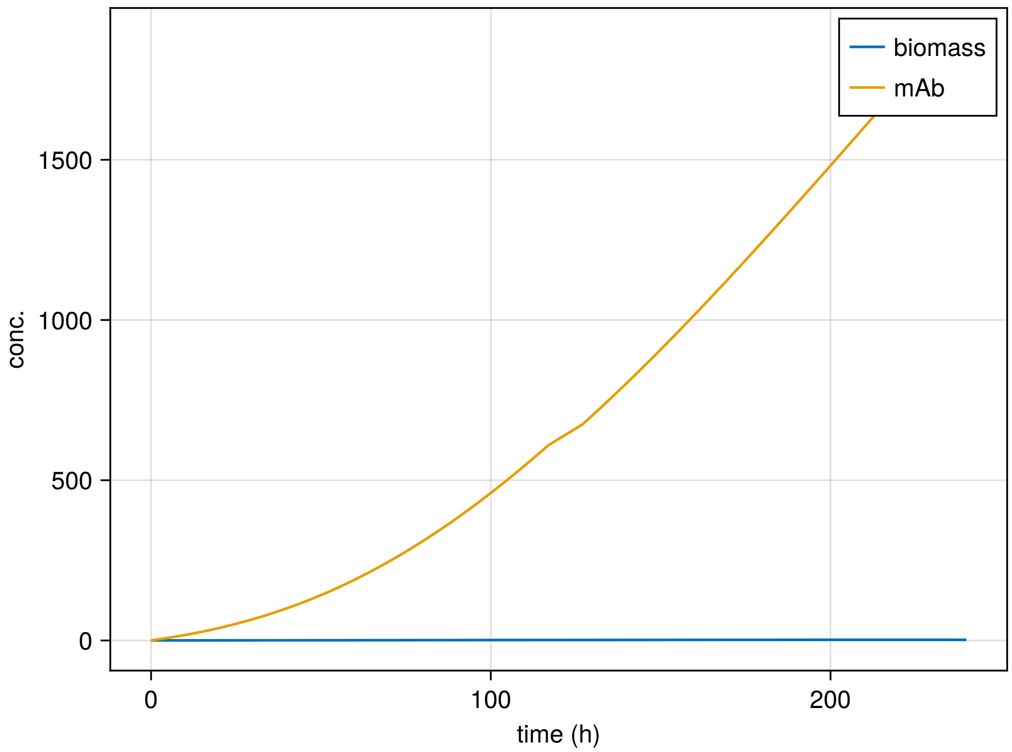

Figure 1: Simulated CHO fed-batch dynamics over 240 h. Top: viable biomass concentration X (g_{DW}/L). Bottom: mAb titer (mg/L). The glucose-triggered feed policy maintains nutrient availability while the Luedeking–Piret non-growth-associated term sustains antibody secretion through the stationary phase.

equations with an algebraic constraint: at metabolic steady state the net rate of production of every intracellular metabolite is zero. Given only the network stoichiometry and bounds on individual reaction rates, the optimal flux distribution consistent with that constraint can be found by linear programming [6].

3.1 Metabolism as a Constraint-Based Problem

The stoichiometry of a metabolic network is encoded in the *stoichiometric matrix* $\mathbf{S} \in \mathbb{R}^{m\times r}$, where m is the number of metabolites and r is the number of reactions. The entry S_{ij} is the stoichiometric coefficient of metabolite i in reaction j: negative if metabolite i is consumed, positive if it is produced, and zero otherwise. The vector $\mathbf{v} \in \mathbb{R}^r$ collects the fluxes of all reactions, in units of, for example, mmol per gram dry weight per hour.

Writing mole balances for each intracellular metabolite in a well-mixed compartment of volume V gives the dynamic equation

$$\frac{d\mathbf{x}}{dt} = \mathbf{S}\,\mathbf{v} - \mu\,\mathbf{x}, \tag{13}$$

where $\mathbf{x}$ is the vector of intracellular metabolite amounts (concentrations in the constant-volume case) and μ is the specific growth rate (dilution by cell growth). Metabolic fluxes operate on the time scale of seconds to minutes, whereas the cell-growth time scale is hours. On the metabolic time scale the dilution term $\mu\,\mathbf{x}$ is negligible relative to $\mathbf{S}\,\mathbf{v}$, and metabolite concentrations reach a pseudo-steady state. Setting $d\mathbf{x}/dt \approx \mathbf{0}$ yields the steady-state mass-balance constraint

$$\mathbf{S}\,\mathbf{v} = \mathbf{0}. \tag{14}$$

The formal derivation of this constraint from open-species mole balances is given in Appendix A.2.

Equation (14) defines a *null space* of feasible flux distributions. For any realistic network with $r > m$ the system is underdetermined; the stoichiometric constraint alone leaves many degrees of freedom. Constraint-based modeling resolves this indeterminacy by (i) imposing thermodynamic and capacity bounds on individual fluxes and (ii) selecting among the feasible solutions via an objective function that reflects cellular physiology [7].

3.2 The FBA Linear Program

Given the steady-state constraint and box bounds on each flux,

$$\begin{aligned} \text{maximize} \quad & \mathbf{c}^\top \mathbf{v} \\ \text{subject to} \quad & \mathbf{S}\,\mathbf{v} = \mathbf{0} \\ & \mathbf{lb} \le \mathbf{v} \le \mathbf{ub}, \end{aligned} \tag{15}$$

where $\mathbf{c} \in \mathbb{R}^r$ is the objective coefficient vector. This is a linear program, solvable in polynomial time for networks of arbitrary size [6, 8].

The lower and upper bounds $\mathbf{lb}$ and $\mathbf{ub}$ encode two types of constraint. Thermodynamic reversibility determines the sign of the lower bound: an irreversible reaction has $lb_j = 0$, whereas a reversible reaction has $lb_j < 0$. Enzyme capacity imposes an upper bound through the maximum velocity $V_{\max,j} = k_{\mathrm{cat},j}\,[E]_j$, where $[E]_j$ is the total enzyme concentration and $k_{\mathrm{cat},j}$ is the turnover number [9].

Exchange and transport reactions. A metabolic network must be coupled to its environment through *exchange reactions* (sometimes called boundary reactions), which represent the import or export of metabolites across the system boundary. By convention, an exchange reaction for metabolite i is written as $\emptyset \to M_i$ so that positive flux denotes import and negative flux denotes export. These pseudo-reactions carry no stoichiometric coefficients for other species and appear as isolated non-zero entries in $\mathbf{S}$. Bounds on exchange reactions encode the composition of the medium: setting $lb_j = 0$ prevents uptake of a nutrient that is absent, while $ub_j = 0$ prevents secretion of a regulated product.

The biological objective. The choice of $\mathbf{c}$ encodes the metabolic goal. Maximizing biomass synthesis – setting $c_j = 1$ for a lumped biomass-assembly reaction – is the canonical choice for growth studies and underpins much of the predictive power of genome-scale FBA [6]. In biotechnological contexts the objective may instead be maximization of a product export flux, reflecting the engineering goal of yield optimization. Regardless of the choice, the linearity of the problem guarantees a unique optimal objective value, although multiple flux vectors may attain it.

3.3 Flux Variability Analysis and Genome-Scale Structure

Because the optimal solution to Eq. (15) is typically not unique, *flux variability analysis* (FVA) characterizes the complete set of optimal solutions. For each reaction j, FVA solves two linear programs: it minimizes and maximizes v_j subject to the FBA constraints plus the requirement that the objective value remain at its optimum. The resulting intervals $[v_j^{\min}, v_j^{\max}]$ identify reactions that are uniquely determined at the optimum (zero-width intervals), reactions that are completely free within the feasible set (wide intervals), and the so-called *blocked* reactions for which the only feasible flux is zero [8].

At the network level, $\mathbf{S}$ encodes the full topology of metabolism. Its null space $\mathcal{N}(\mathbf{S})$ – the set of all flux vectors satisfying $\mathbf{S}\,\mathbf{v} = \mathbf{0}$ – has dimension $r - \mathrm{rank}(\mathbf{S})$. The singular-value decomposition $\mathbf{S} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ reveals this structure directly: the columns of $\mathbf{V}$ corresponding to zero singular values span $\mathcal{N}(\mathbf{S})$, while the remaining columns of $\mathbf{V}$ span the row space. In genome-scale models (thousands of reactions, hundreds of metabolites) $\mathrm{rank}(\mathbf{S}) \ll r$, so the null space is high-dimensional and the feasible solution space is correspondingly large, motivating the use of physiological objective functions to select biologically meaningful flux distributions [5, 7].

3.4 Worked Example: The Urea Cycle

To illustrate FBA concretely, we apply it to a curated model of the urea cycle in HL-60 cells. The network comprises 5 enzymatic reactions and 14 exchange reactions, coupling 18 intracellular metabolites ($m = 18$, $r = 19$, so $\mathbf{S} \in \mathbb{R}^{18\times 19}$).

Reactions. The four canonical urea-cycle enzymes are represented:

- **v1** (argininosuccinate synthetase, EC 6.3.4.5): ATP + Citrulline + Aspartate $\to$ AMP + Diphosphate + Argininosuccinate;
- **v2** (argininosuccinate lyase, EC 4.3.2.1): Argininosuccinate $\to$ Fumarate + Arginine;
- **v3** (arginase I, EC 3.5.3.1): Arginine + H_2O $\to$ Ornithine + Urea;
- **v4** (ornithine transcarbamylase, EC 2.1.3.3): Carbamoyl phosphate+Ornithine $\to$ Orthophosphate+ Citrulline.

A nitric oxide synthase branch is also included:

- **v5** (NOS, EC 1.15.13.39): $2\,\text{Arginine} + 4\,O_2 + 3\,\text{NADPH} + 3\,H^+ \rightarrow 2\,\text{NO} + 2\,\text{Citrulline} + 3\,\text{NADP}^+ + 4\,H_2O$.

Exchange reactions b1–b14 allow each metabolite to enter or leave the system at bounded rates.

Flux bounds. Reversibility is assigned from the standard Gibbs free energy of reaction evaluated by eQuilibrator: reactions with $\Delta G^{\circ\prime} < -10\,\text{kJ/mol}$ are treated as irreversible ($lb_j = 0$); the remainder are reversible. Capacity upper bounds are set as $V_{\max,j} = k_{\text{cat},j}\,[E]_0$ with $[E]_0 = 0.01\,\text{mmol gDW}^{-1}$ and turnover numbers from BRENDA. The rate-limiting step in this parameterization is v2 (argininosuccinate lyase), whose $k_{\text{cat}} = 3.28\,\text{s}^{-1}$ gives $V_{\max,\text{v2}} = 0.0328\,\text{mmol gDW}^{-1}\,\text{h}^{-1}$ — the smallest of the five enzymatic capacity bounds (v1: 0.100, v3: 1.9, v4: 4.1, v5: 0.100) and therefore the unique network bottleneck.

Objective and sign convention. The biological goal is to maximize urea export. Exchange reaction b4 is defined with the import-positive convention ($\emptyset \rightarrow \text{Urea}$), so urea export corresponds to *negative* flux through b4. To drive b4 as negative as possible under a maximization objective, the objective coefficient is set to $c_{\text{b4}} = -1$: maximizing $\mathbf{c}^\top \mathbf{v}$ then minimizes v_{b4}, which is equivalent to maximizing the rate of urea export.

Optimal flux distribution. The linear program (15) was solved with the HiGHS solver via JuMP. At the optimum the entire urea-cycle backbone (v1–v4) operates at exactly $0.0328\,\text{mmol gDW}^{-1}\,\text{h}^{-1}$ – the capacity imposed by v2 – and the NO synthase branch (v5) carries zero flux, consistent with arginine being fully committed to urea production. The corresponding exchange fluxes shuttle carbamoyl phosphate, aspartate, and ATP into the cycle while exporting urea, fumarate, AMP, and orthophosphate. The optimal urea-export flux magnitude is $0.0328\,\text{mmol gDW}^{-1}\,\text{h}^{-1}$, confirming that the cycle rate is limited by argininosuccinate lyase. The full flux distribution is shown in Figure 2.

4 Models of Gene Expression

Gene expression is controlled not by individual promoters in isolation but by networks of interacting regulatory proteins, small molecules, and nucleic acid sequences. Three complementary frameworks span the range from the shape of a single regulatory binding curve to the behavior of multi-gene circuits and whole-cell resource allocation: (i) Hill-type input functions that describe how transcription factor occupancy depends on effector concentration, (ii) ordinary differential equation (ODE) network-motif models that reveal how recurring connectivity patterns give rise to specific dynamical behaviors, and (iii) cybernetic models that treat the cell as an optimizer allocating limited biosynthetic resources among competing metabolic pathways. Together these frameworks provide a mechanistic vocabulary for understanding and engineering gene regulatory circuits.

4.1 Regulatory Input Functions: the Hill Equation

The probability that a promoter is occupied by a transcription factor—and therefore the fractional transcriptional activity of the target gene—is the central input function of gene-circuit models. For a transcriptional activator X that binds cooperatively to n sites on a promoter with dissociation

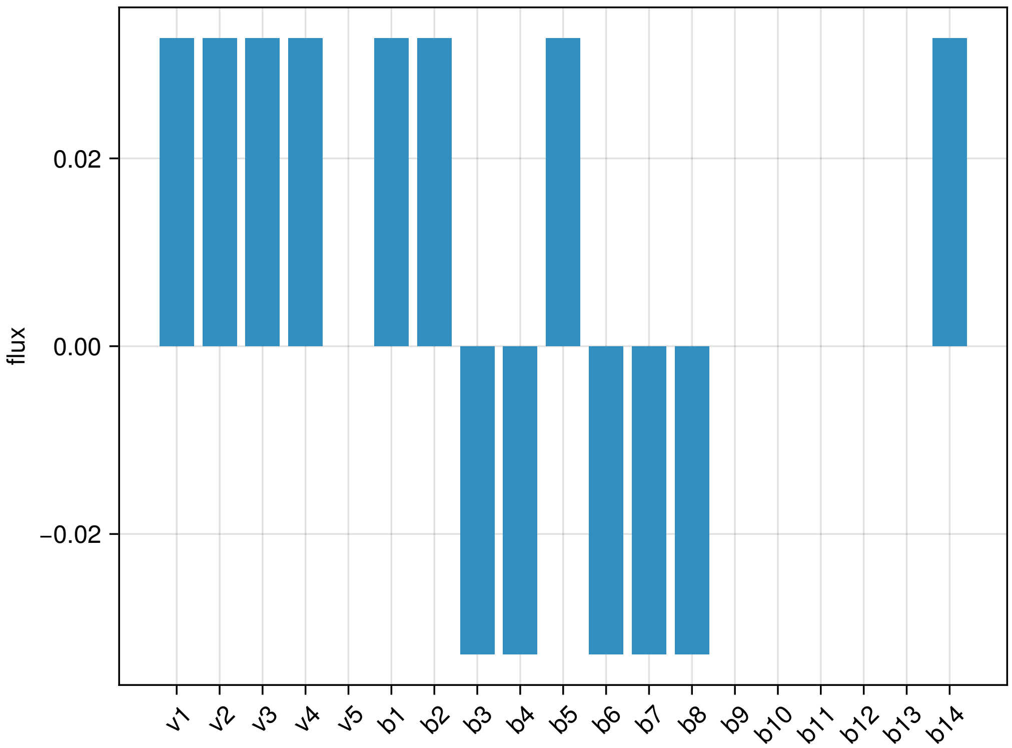

Figure 2: Optimal flux distribution for the urea-cycle FBA model. Bar heights give the flux v_j ($\mathrm{mmol\,gDW^{-1}\,h^{-1}}$) at the solution of linear program (15) with the urea-export objective. Enzymatic reactions v1–v4 all saturate at $0.0328\,\mathrm{mmol\,gDW^{-1}\,h^{-1}}$, the capacity of argininosuccinate lyase (v2). The NO synthase branch (v5) carries zero flux. Exchange reactions (b1–b14) with negative bars represent net export under the import-positive convention; b4 (urea) shows the maximized export flux of $-0.0328\,\mathrm{mmol\,gDW^{-1}\,h^{-1}}$.

constant K (in concentration units), equilibrium statistical mechanics yields the activating Hill function [10]:

$$f_{\rm act}(x) = \frac{x^n}{K^n + x^n}, \tag{16}$$

where x denotes the concentration of X. The function rises from 0 to 1 as x increases, with $f_{\rm act}(K) = 1/2$; thus K is the half-maximal activation concentration. For a repressor that occludes the same n binding sites the complementary repressing Hill function is

$$f_{\rm rep}(x) = \frac{K^n}{K^n + x^n} = 1 - f_{\rm act}(x). \tag{17}$$

The Hill coefficient n measures the steepness (cooperativity) of the response. When $n = 1$ the binding curve is a simple rectangular hyperbola; for $n > 1$ the curve becomes progressively more switch-like, approaching a step function as $n \to \infty$. Physically, positive cooperativity ($n > 1$) arises when the binding of one ligand increases the affinity for subsequent ligands—as in allosteric proteins—or when n molecules must bind simultaneously before the complex is active. In the context of gene regulation the effective cooperativity is often set by the multimerization state of the transcription factor or by the number of operator sites. A full derivation of the Hill form from equilibrium binding of n cooperative ligands is given in Appendix Section A.3.

Because Eqs. (16) and (17) are bounded between zero and one, any production rate can be written as a maximal rate multiplied by a Hill function. Multiple regulators are combined multiplicatively (AND logic) or additively (OR logic), making Hill functions the universal building block of gene-circuit ODEs. In the motif.jl implementation used throughout this section the Hill functions are defined as `hill_act(x; K, n)` $= x^n/(K^n+x^n)$ and `hill_rep(x; K, n)` $= K^n/(K^n+x^n)$, with default cooperativity $n = 2$.

4.2 Network Motifs as Design Principles

Analysis of the *Escherichia coli* transcriptional regulation network revealed that a small number of three- and four-node connectivity patterns occur far more often than expected by chance [11]. These "network motifs" are not merely statistical curiosities; each recurring pattern confers a specific dynamical function that can be understood analytically and verified computationally [10, 12]. Figure 3 illustrates the four motifs developed in this section: negative autoregulation, the coherent type-1 feed-forward loop, the incoherent type-1 feed-forward loop, and a three-component negative-feedback oscillator. A compact catalog of these and related motifs can be found in Shoval and Alon [13].

Negative Autoregulation

The simplest circuit with a non-trivial design function is one in which a transcription factor Y represses its own gene. With Hill repression and first-order degradation, the ODE governing Y is

$$\frac{dY}{dt} = \beta_{\rm nar}\, f_{\rm rep}(Y;\, K,\, n) - \alpha\, Y, \tag{18}$$

where $\beta_{\rm nar}$ is the maximal synthesis rate and α is the degradation rate constant. The steady state Y^* satisfies $\beta_{\rm nar}\, f_{\rm rep}(Y^*;\, K,\, n) = \alpha\, Y^*$, so

$$\beta_{\rm nar} = \frac{\alpha\, Y^*}{f_{\rm rep}(Y^*;\, K,\, n)}. \tag{19}$$

To isolate the effect of feedback, the unregulated (simple) comparator has the constant production rate $\beta_{\text{simple}} = \alpha\, Y^*$, giving identical steady states to both circuits. Early in the response, when $Y \approx 0$, $f_{\text{rep}}(0;\, K,\, n) = 1$, so the NAR circuit initially produces at the elevated rate $\beta_{\text{nar}} > \beta_{\text{simple}}$; as Y rises above K the repressor begins to shut itself off, braking synthesis just as the target concentration is approached. The net result is a faster rise to the common steady state: the half-response time of NAR is strictly less than that of the unregulated circuit [10]. Panel A of Figure 3 confirms this behavior. Beyond speed, negative autoregulation reduces cell-to-cell variability in protein level: because the circuit corrects perturbations about Y^*, stochastic fluctuations are attenuated relative to the unregulated case.

Coherent Type-1 Feed-Forward Loop (C1-FFL): Persistence Detector

A feed-forward loop (FFL) adds a second regulatory arm to a linear cascade. In the coherent type-1 FFL (C1-FFL) the master regulator X activates both the intermediate Y and the output Z, while Y also activates Z (all interactions are activating) [14]. With an AND gate at Z (both X and Y must be sufficiently high to drive Z production), the governing ODEs are

$$\frac{dY}{dt} = \beta_Y\, f_{\text{act}}(X;\, K_{XY},\, n) - \alpha_Y\, Y, \tag{20}$$

$$\frac{dZ}{dt} = \beta_Z\, f_{\text{act}}(X;\, K_{XZ},\, n)\, f_{\text{act}}(Y;\, K_{YZ},\, n) - \alpha_Z\, Z. \tag{21}$$

The design function follows from a timing argument detailed in Appendix Section A.4: a transient pulse of X is filtered because Y cannot accumulate fast enough (on the time scale $1/\alpha_Y$) to open the AND gate before X turns off again; a sustained step of X, however, keeps the direct arm open long enough for Y to rise above K_{YZ}, at which point Z turns on after a sign-sensitive delay of approximately $1/\alpha_Y$. Panel B of Figure 3 shows this behavior: a brief input pulse leaves Z essentially unaffected, while the subsequent sustained step elicits a delayed but robust Z response. The C1-FFL thus acts as a persistence detector—or equivalently as a noise filter that rejects transient signals while passing sustained signals.

Incoherent Type-1 Feed-Forward Loop (I1-FFL): Pulse Generator

In the incoherent type-1 FFL (I1-FFL) the regulatory signs are mixed: X activates both Y and Z, but Y represses Z. The ODEs are

$$\frac{dY}{dt} = \beta_Y\, f_{\text{act}}(X;\, K_{XY},\, n) - \alpha_Y\, Y, \tag{22}$$

$$\frac{dZ}{dt} = \beta_Z\, f_{\text{act}}(X;\, K_{XZ},\, n)\, f_{\text{rep}}(Y;\, K_{YZ},\, n) - \alpha_Z\, Z. \tag{23}$$

When X steps on, the direct activating arm immediately drives Z upward; simultaneously, X activates the slower indirect arm, accumulating Y on the time scale $1/\alpha_Y$. Once Y is sufficiently high, its repression of Z overrides the direct activation, pulling Z back toward a lower steady state. The result is a transient pulse in Z: the peak output exceeds the eventual steady state by a factor that depends on the ratio of the direct and indirect pathway strengths (Appendix Section A.4). Panel C of Figure 3 shows the pulse. A particularly important property of the I1-FFL is fold-change detection: the peak of Z is determined by the fold-change in X, not its absolute level [15]. This scale-invariance allows the circuit to respond consistently to signals encountered against widely varying backgrounds.

Negative-Feedback Oscillator (Goodwin)

A three-variable negative-feedback loop, originally analyzed by Goodwin [10], can generate sustained oscillations when the Hill coefficient is sufficiently high. The ODE system is

$$\frac{dX}{dt} = a\, f_{\mathrm{rep}}(Z;\, K,\, n) - b\, X, \tag{24}$$
$$\frac{dY}{dt} = c\, X - d\, Y, \tag{25}$$
$$\frac{dZ}{dt} = e\, Y - f\, Z, \tag{26}$$

where the repressor Z feeds back to suppress X through a Hill repression term. The three-stage cascade $X \to Y \to Z \dashv X$ introduces a phase lag around the loop; when the nonlinearity is sharp enough ($n \geq 9$) the fixed point becomes unstable and the system settles onto a limit cycle. Panel D of Figure 3 shows sustained oscillations of X, Y, and Z with six peaks across the simulation window, consistent with a period of roughly 15–20 time units. Biologically, this architecture appears in circadian clocks (where Z is the repressor protein PER/CRY in the mammalian clock network) and in certain cell-cycle controllers.

A note on biophysical promoter models. The Hill functions developed here capture regulatory logic at the level of effective binding equilibria. More granular biophysical descriptions, including effective-biophysical models that explicitly resolve transcription and translation elongation rates [16] and structured-promoter biosensor models that embed sequence-level parameters, provide the next tier of mechanistic detail; they are cited here but not derived in this chapter.

4.3 Cybernetic Resource Allocation: Diauxic Growth

The gene-circuit models of Section 4.2 treat a single regulatory node or a small defined circuit. At the whole-cell scale, hundreds of metabolic pathways compete for a shared pool of ribosomes, transcription machinery, and energy. Cybernetic models, introduced by Kompala and Ramkrishna [17, 18], treat these allocation decisions as the outcome of optimal resource distribution: the cell is assumed to invest enzyme synthesis in whichever pathway yields the greatest current return, generating emergent diauxic growth behavior without prescribing substrate preference explicitly.

The Matching-Law Cybernetic Model

Consider a culture growing on two substrates S_1 (glucose) and S_2 (xylose), each consumed by a dedicated enzymatic pathway E_i with enzyme-specific growth rate

$$rG_i = \mu_{\max,i}\, \frac{E_i}{E_{\max,i}}\, \frac{S_i}{K_i + S_i}, \qquad i = 1, 2, \tag{27}$$

where $\mu_{\max,i}$ is the maximum pathway-specific growth rate and K_i is the Monod half-saturation constant. The enzyme concentration E_i modulates the effective pathway capacity.

Cybernetic control is embodied in two variables derived from the relative returns rG_i:

$$u_i = \frac{rG_i}{\sum_j rG_j} \tag{28}$$

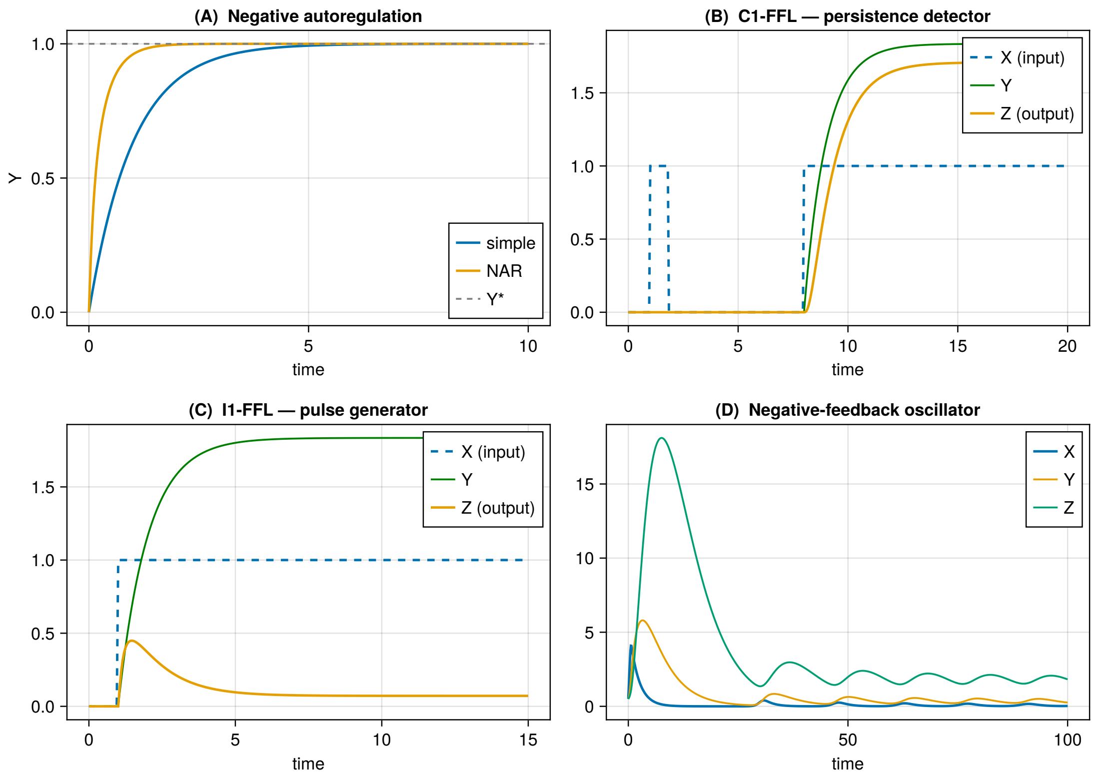

Figure 3: Four canonical network motifs simulated with Hill input functions. (A) Negative autoregulation (NAR) versus simple regulation: both circuits share the same steady state Y^* (dashed line) but NAR reaches half-maximal response faster. (B) Coherent type-1 FFL (AND gate): a brief input pulse (left of $t = 2$) is filtered and Z remains near zero; a sustained step (beginning at $t = 8$) allows Y to accumulate and Z turns on after a delay of $\approx 1/\alpha_Y$. (C) Incoherent type-1 FFL: Z exhibits a pulse upon the step increase in X, with peak exceeding the eventual steady state by more than sixfold. (D) Goodwin-style three-variable negative-feedback oscillator with Hill coefficient $n = 10$; sustained limit-cycle oscillations persist throughout the simulation.

allocates the *enzyme synthesis* rate according to the matching law: each pathway receives a fraction of the total ribosomal investment equal to its fractional contribution to the instantaneous growth rate. Separately,

$$v_i = \frac{rG_i}{\max_j rG_j} \tag{29}$$

modulates *enzyme activity*: existing enzyme molecules are deployed most fully in the highest-return pathway, and partially downregulated in others. The derivation and biological motivation of these allocation rules as a consequence of maximizing the instantaneous growth rate are given in Appendix Section A.5. The same allocation idea can also be cast as a linear program that chooses the single best pathway at each instant (the Kompala-LP reframing), but the matching-law variant—implemented here—allows graded co-utilization of pathways.

The net specific growth rate is

$$\mu = \sum_i rG_i \, v_i, \tag{30}$$

and the governing ODEs for the five-state system (S_1, S_2, E_1, E_2, C), where C is biomass, are

$$\frac{dS_i}{dt} = -\frac{1}{Y_i} rG_i \, v_i \, C, \tag{31}$$

$$\frac{dE_i}{dt} = r_{E_i} \, u_i - (\mu + \beta_i) \, E_i, \tag{32}$$

$$\frac{dC}{dt} = (\mu - k_d) \, C, \tag{33}$$

where Y_i is the biomass yield on substrate i, $r_{E_i} = 1/\tau_i$ is the maximal enzyme synthesis rate (active only when $S_i > 0$), β_i is the enzyme degradation rate, and k_d is the maintenance death rate. The enzyme balance (Eq. 32) shows the matching law directly: synthesis is invested in enzyme i at rate $r_{E_i} \, u_i$, so when $rG_1 \gg rG_2$ nearly all new enzyme is directed to pathway 1.

Diauxic Growth Dynamics

Figure 4 shows biomass C and both substrate concentrations simulated over 20 h starting from $S_1 = 0.5$ g/L (glucose), $S_2 = 2.5$ g/L (xylose), and a low initial biomass of 4×10^{-3} $\mathrm{g_{DW}}$/L. The matching law concentrates enzyme synthesis on glucose because $\mu_{\max,1} = 1.08$ h^{-1} exceeds $\mu_{\max,2} = 0.82$ h^{-1} and $K_1 = 0.01$ g/L is much smaller than $K_2 = 0.20$ g/L, making glucose the higher-return substrate. Consequently S_1 is exhausted before S_2 begins to fall appreciably. After glucose depletion, $rG_1 \to 0$ and the matching law redirects synthesis toward the xylose pathway; $u_2 \to 1$ and enzyme E_2 accumulates, so S_2 consumption resumes after a brief lag. Biomass grows continuously, with a distinct dip in growth rate between the two substrate phases. The model reproduces qualitatively the classic diauxic phenotype reported experimentally by Kompala et al. [18] and predicted analytically by Kompala et al. [17].

5 Deep Time-Series Models

5.1 Motivation: Data-Driven Sequence Models for Bioprocesses

Mechanistic models of the kind developed in Section 2 rest on the identification of every kinetically significant reaction, the measurement of the corresponding parameters, and the verification of structural assumptions through targeted experiments. Each of these steps is expensive. When the network is incompletely known, when kinetic parameters cannot be reliably estimated, or when the

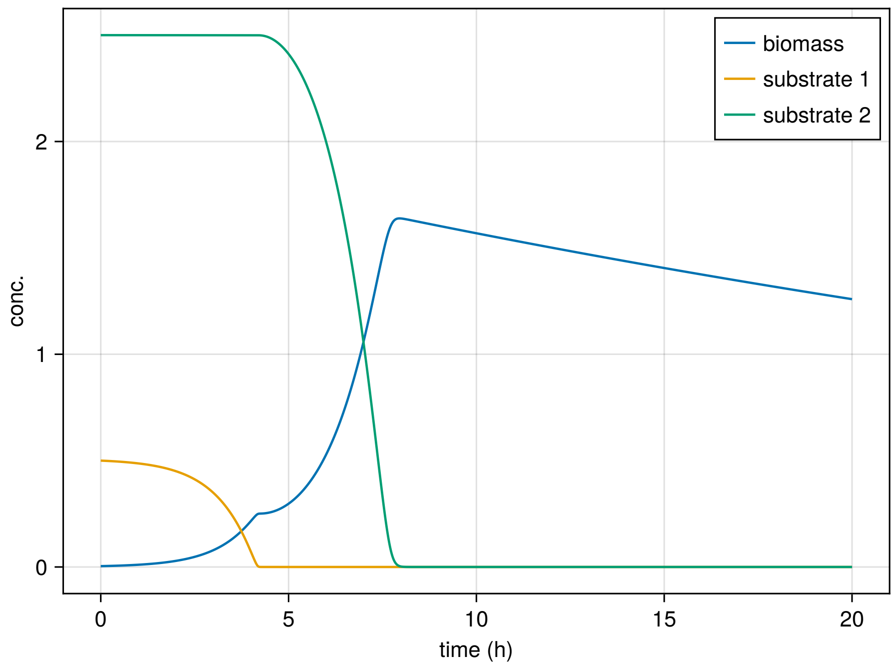

Figure 4: Cybernetic model simulation of diauxic growth on glucose (S_1, g/L) and xylose (S_2, g/L). Biomass C grows exponentially during glucose consumption, pauses briefly at the diauxic lag, then resumes as the matching-law control variables redirect enzyme synthesis to the xylose pathway. Both substrates are fully depleted by $t \approx 12$ h.

available experimental budget precludes full mechanistic closure, a data-driven alternative becomes attractive. Deep sequence models infer temporal structure directly from measured trajectories without committing to a mechanistic hypothesis.

The trade-off is familiar from statistical learning theory. A fully parametric mechanistic model has low variance (the solution space is tightly constrained by mass-action stoichiometry and thermodynamics) but potentially high bias if the assumed structure is wrong. A flexible sequence model has lower bias but higher variance: it requires substantially more training data to avoid overfitting, and its predictions outside the training distribution are unreliable. For bioprocesses this boundary is consequential. Mechanistic models extrapolate gracefully to process conditions not seen during parameter estimation, provided the rate-law forms are correct. Data-driven models interpolate accurately within the training envelope but can fail catastrophically when asked to extrapolate to genuinely new conditions.

Despite these limitations, deep sequence models occupy a legitimate role in bioprocess engineering. Process development data accumulated across many batches, combined with digital-twin simulation (see Section 2), can provide the volume and diversity of trajectories needed to train reliable forecasters. The models developed in this section use synthetic training data generated by the CHO mAb mechanistic model of Section 2, making the comparison clean and the ground truth exact while illustrating the methodology applicable to real process databases.

5.2 Recurrent Architectures: RNN, LSTM, and Structured State Spaces

5.2.1 Elman Recurrent Neural Network

The Elman recurrent neural network (RNN) maintains a hidden state $\mathbf{h}_t \in \mathbb{R}^{d_h}$ that summarizes the history of the input sequence $\mathbf{u}_1, \ldots, \mathbf{u}_t$. The update rule is

$$\mathbf{h}_t = \phi(W_{hh}\,\mathbf{h}_{t-1} + W_{uh}\,\mathbf{u}_t + \mathbf{b}_h)\,, \tag{34}$$

where $W_{hh} \in \mathbb{R}^{d_h \times d_h}$ and $W_{uh} \in \mathbb{R}^{d_h \times d_u}$ are learnable weight matrices, $\mathbf{b}_h$ is a bias vector, and $\phi(\cdot)$ is an element-wise nonlinearity such as tanh. An output $\hat{\mathbf{y}}_t = W_{yh}\mathbf{h}_t$ is read from the final hidden state.

Training is performed by backpropagation through time (BPTT): the recurrence is unrolled over the sequence length T, and gradients are propagated backward through all T copies of the weight matrices. Because the same W_{hh} appears at every step, the gradient of the loss $\mathcal{L}$ with respect to parameters at step t involves products of the Jacobian $\partial\mathbf{h}_k/\partial\mathbf{h}_{k-1}$ evaluated at $T-t$ consecutive time steps. When the dominant singular value of this Jacobian is less than one, these products decay exponentially with sequence length, causing gradients to vanish; when it exceeds one, they grow without bound, causing gradients to explode. The vanishing-gradient problem renders plain RNNs unable to learn dependencies that span more than a few dozen time steps. A full derivation of BPTT and the vanishing-gradient mechanism is given in Appendix A.6.

5.2.2 Long Short-Term Memory

The long short-term memory (LSTM) network introduced by Hochreiter and Schmidhuber [19] resolves the vanishing-gradient problem by adding a separate cell state $\mathbf{c}_t$ whose update path is protected by multiplicative gates. Three sigmoid-activated gates control information flow: an input gate $\mathbf{i}_t$, a forget gate $\mathbf{f}_t$, and an output gate $\mathbf{o}_t$. Together with a candidate cell update $\tilde{\mathbf{c}}_t$, the LSTM equations are

$$\mathbf{i}_t = \sigma(W_i\,[\mathbf{h}_{t-1};\mathbf{u}_t] + \mathbf{b}_i)\,, \tag{35}$$
$$\mathbf{f}_t = \sigma(W_f\,[\mathbf{h}_{t-1};\mathbf{u}_t] + \mathbf{b}_f)\,, \tag{36}$$
$$\mathbf{o}_t = \sigma(W_o\,[\mathbf{h}_{t-1};\mathbf{u}_t] + \mathbf{b}_o)\,, \tag{37}$$
$$\tilde{\mathbf{c}}_t = \tanh(W_c\,[\mathbf{h}_{t-1};\mathbf{u}_t] + \mathbf{b}_c)\,, \tag{38}$$
$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \tag{39}$$
$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \tag{40}$$

where $[\mathbf{h}_{t-1};\mathbf{u}_t]$ denotes concatenation, σ is the logistic sigmoid, and $\odot$ is element-wise multiplication. The forget gate $\mathbf{f}_t$ can be close to one, allowing the cell state $\mathbf{c}_t$ to carry information across many steps without multiplicative decay, which is the structural mechanism that avoids vanishing gradients. A derivation of the gate equations and an analysis of the gradient flow through the cell state is given in Appendix A.7.

The implementation used here (see `code/deeplearning/lstm.jl`) is a single Flux `LSTM` layer with hidden size 32, followed by a `Dense` readout that maps the final hidden state to a flattened multi-step forecast of shape ($N_{\text{states}} \times T_{\text{horizon}}$). The model is trained with Adam at a learning rate of 10^{-3} for 200 epochs on sliding-window pairs with lookback $L = 24$ and horizon $H = 12$.

5.2.3 Structured State-Space Models (S4 / HiPPO-LegS)

Structured state-space models (S4) [20] recast the sequence model as a continuous linear dynamical system

$$\frac{d\mathbf{x}}{dt} = A\,\mathbf{x}(t) + B\,\mathbf{u}(t), \qquad \mathbf{y}(t) = C\,\mathbf{x}(t) + D\,\mathbf{u}(t), \tag{41}$$

where $\mathbf{x}(t) \in \mathbb{R}^{N_{\text{lat}}}$ is a latent state, $\mathbf{u}(t)$ is the input, and (A, B, C, D) are parameter matrices. For discrete-time sequences the continuous system is converted to a recurrence via bilinear (Tustin) discretization at step Δt:

$$\bar{A} = \left(I - \tfrac{\Delta t}{2} A\right)^{-1} \left(I + \tfrac{\Delta t}{2} A\right), \qquad \bar{B} = \left(I - \tfrac{\Delta t}{2} A\right)^{-1} \Delta t\, B, \tag{42}$$

yielding the discrete recurrence $\mathbf{x}_k = \bar{A}\,\mathbf{x}_{k-1} + \bar{B}\,\mathbf{u}_k$ with readout $\mathbf{y}_k = C\,\mathbf{x}_k + D\,\mathbf{u}_k$.

The key innovation of S4 is the HiPPO-LegS initialization, which sets A and B to a specific matrix pair designed so that the latent state $\mathbf{x}(t)$ optimally compresses the history of $\mathbf{u}$ into a polynomial basis on the time interval $[0, t]$ weighted by the Legendre measure. For a single-channel system of order h, the HiPPO-LegS matrices are

$$A_{nk} = -\begin{cases} \sqrt{(2n+1)(2k+1)} & n > k, \\ n+1 & n = k, \\ 0 & n < k, \end{cases} \qquad b_n = \sqrt{2n+1}, \quad n, k = 1, \ldots, h. \tag{43}$$

A full derivation of the bilinear discretization and a note on the HiPPO-LegS construction are given in Appendix A.8.

The implementation used here (`code/deeplearning/s4.jl`) applies one independent h-dimensional LegS filter per input channel in a block-diagonal arrangement, giving total latent dimension $N_{\text{lat}} = h \times N_{\text{states}}$ with $h = 16$. Critically, the discrete operators $\bar{A}$ and $\bar{B}$ are *fixed* at their HiPPO-LegS values and are never updated by gradient descent; only the readout C and the feedforward D are trainable. This is the principled initialization-fixed S4 training recipe: the structured state matrix provides a universal long-range memory, and only the linear projection from that memory to the target is learned. The model is trained with Adam at a learning rate of 10^{-2} for 300 epochs on the same sliding windows as the LSTM (lookback $L = 24$, horizon $H = 12$, seed 42).

5.3 Worked Example: S4 vs LSTM on a CHO mAb Bioreactor

To illustrate both architectures in a realistic bioprocess setting, we trained the LSTM and S4 forecasters on synthetic CHO mAb batch trajectories generated by the mechanistic model of Section 2. This setup provides an exact ground truth and a fully controlled experimental design, making the comparison reproducible without requiring access to proprietary industrial data.

Data and protocol. Trajectories of the seven process states — culture volume V, viable cell density X, glucose [Glc], glutamine [Gln], lactate [Lac], ammonium [Amm], and monoclonal antibody [mAb] — were generated by forward integration of the CHO kinetic model. Sliding windows were constructed with lookback $L = 24$ and forecast horizon $H = 12$ time steps. States were normalized to zero mean and unit variance; the training and test sets were constructed with a fixed split and random seed 42 to ensure identical conditions for both models.

Results. Both models learn the overall trajectory shapes, but the per-state root-mean-square errors on the held-out test windows differ substantially. Table 1 summarizes the raw-scale RMSE for each state.

Table 1: Per-state test RMSE for the LSTM and S4 forecasters on the CHO mAb bioreactor. Training data are synthetic trajectories from the mechanistic model of Section 2. Lower is better.

State	Units	LSTM RMSE	S4 RMSE
V	L	0.0146	0.0096
X	gDW/L	0.0335	0.0284
[Glc]	mM	4.976	3.936
[Gln]	mM	1.640	1.298
[Lac]	mM	2.484	0.532
[Amm]	mM	0.4433	0.09520
[mAb]	mg/L	162.1	34.63

In this demonstration S4 achieved lower RMSE than the LSTM on all seven states. The largest relative improvement is on [mAb] (LSTM: 162.1 mg/L vs. S4: 34.63 mg/L) and on [Amm] (LSTM: 0.4433 mM vs. S4: 0.09520 mM), both of which display smooth, monotonically evolving profiles over the batch. The HiPPO-LegS memory structure appears well-suited to such long-range monotone trends, as it encodes the entire lookback window into a polynomial projection before the (linear) readout is applied.

Figures 5 and 6 show representative 12-step-ahead forecast panels for the LSTM and S4 models, respectively. Figure 7 presents the head-to-head RMSE comparison across all seven states.

Interpretation. This comparison is a single demonstration on one synthetic dataset and one set of hyperparameters. It is not evidence of a universal advantage of S4 over LSTM across bioprocess forecasting tasks. The LSTM is a more general architecture with nonlinear gating; the S4 variant used here exploits a particular linear structure that may match the smooth, nearly exponential

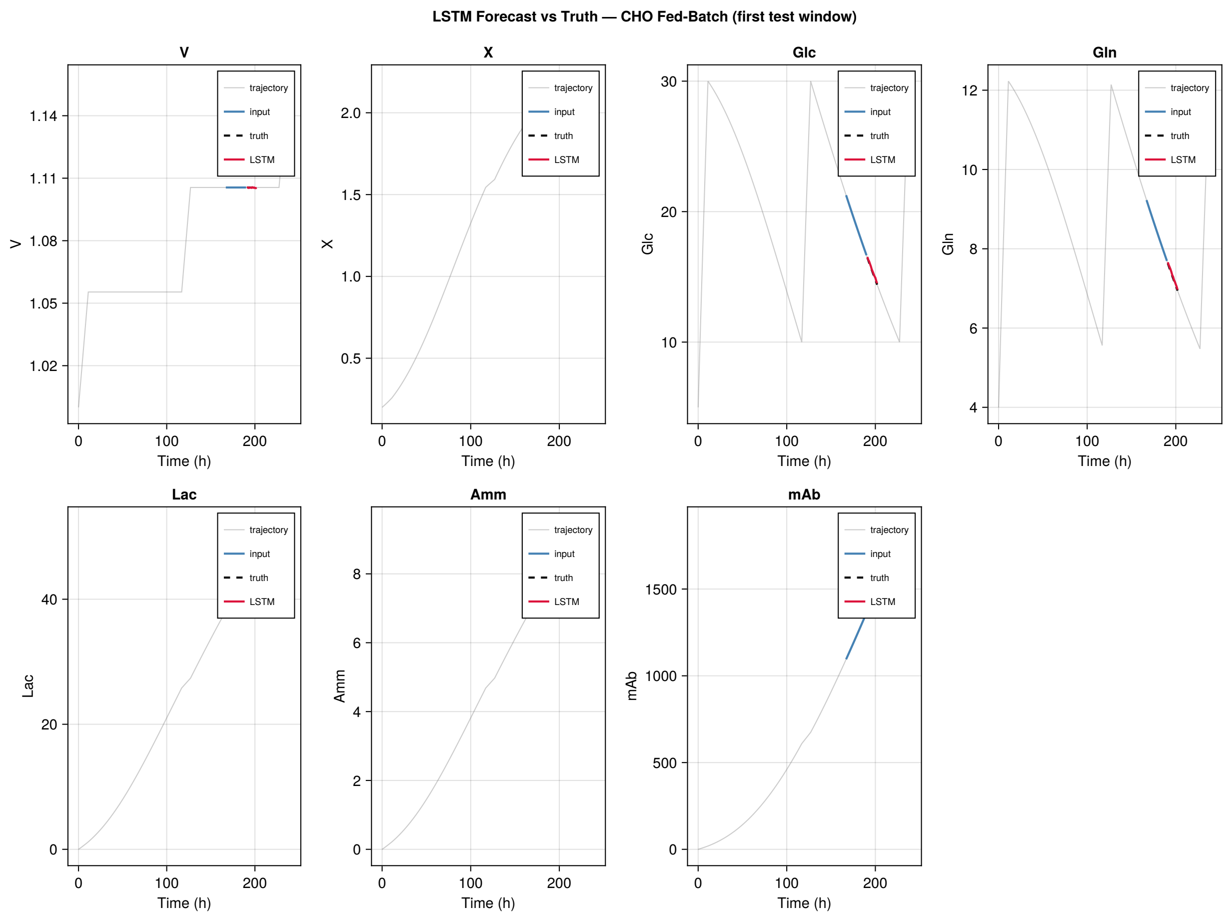

Figure 5: LSTM 12-step-ahead forecast on the held-out CHO mAb test set (first test window). Each panel shows the 24-step input window (blue), the 12-step ground-truth horizon (dashed), and the LSTM prediction (red) for one of the seven process states. Lookback $L = 24$, horizon $H = 12$, hidden size 32.

profiles of batch bioreactor states. On datasets with strong nonlinear transients or irregular sampling, an LSTM or its variants may be preferable. The appropriate choice depends on dataset size, sequence length, and the qualitative character of the process dynamics.

5.4 Transformers and Self-Attention

Transformer architectures [21] compute pairwise attention scores between every pair of positions in the input sequence, allowing the model to relate distant time steps in a single operation rather than propagating information step by step through a recurrence. For a sequence of length T and embedding dimension d_{model}, the scaled dot-product attention mechanism computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{44}$$

where the query matrix Q, key matrix K, and value matrix V are linear projections of the input, and d_k is the key dimension. Multi-head attention runs H_{heads} such operations in parallel

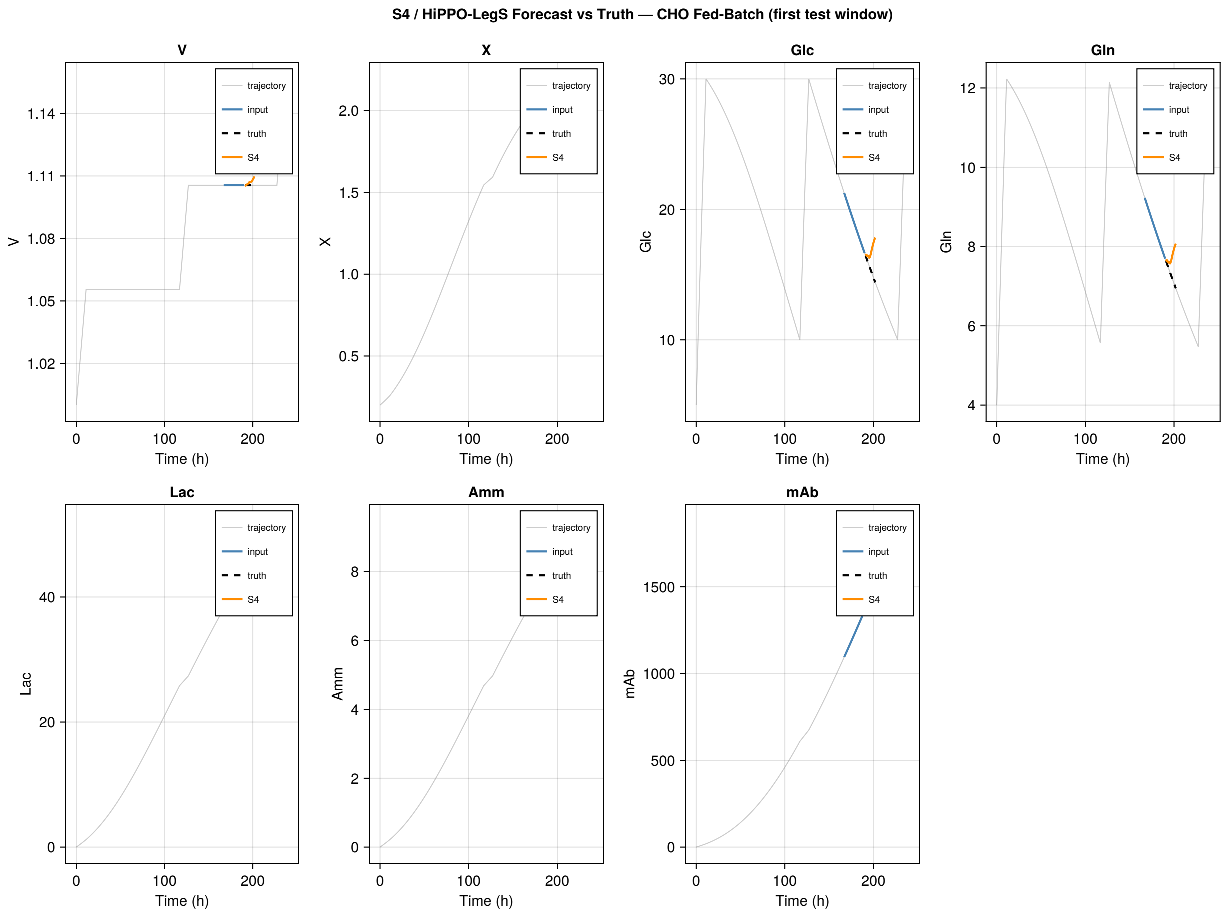

Figure 6: S4/HiPPO-LegS 12-step-ahead forecast on the same held-out CHO mAb test set as Figure 5 (first test window). Each panel shows the 24-step input window (blue), the 12-step ground-truth horizon (dashed), and the S4 prediction (orange). The state-space dynamics $(\bar{A}, \bar{B})$ are fixed at their HiPPO-LegS values; only the readout C and feedforward D are trained. Per-channel LegS order $h = 16$.

and concatenates the results, allowing the model to attend to different representational subspaces simultaneously.

Transformers have achieved state-of-the-art performance on long-horizon time-series forecasting tasks in many domains, and temporal variants (e.g., the Informer, Autoformer, and PatchTST families) are increasingly applied to industrial process data. Their primary advantage over recurrent architectures is parallelization during training, since the attention computation does not depend on the sequential order of computation. Their primary disadvantage is the $O(T^2)$ memory cost of the attention matrix, which constrains sequence length in practice and requires specialized approximations for very long windows.

For the bioprocess forecasting tasks considered in this chapter — batch sequences of order 10^2 to 10^3 time steps — either recurrent models (LSTM, S4) or Transformer-based models are computationally feasible. The Transformer family becomes particularly attractive when large pre-trained sequence models can be fine-tuned on process data, amortizing the high data cost of training from scratch. A detailed implementation of Transformer-based bioprocess forecasting is beyond the

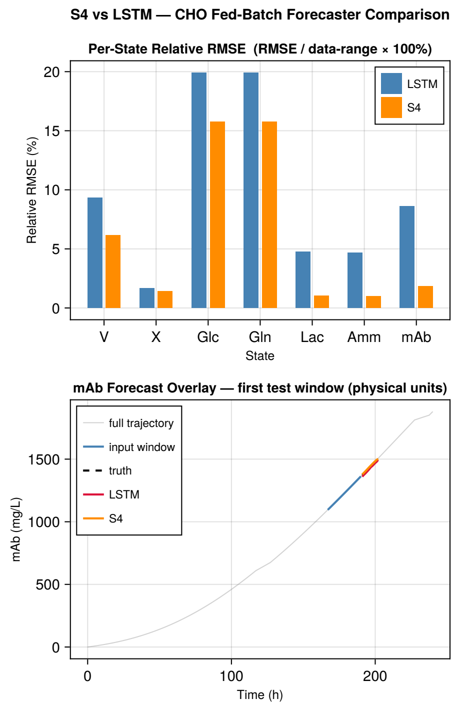

Figure 7: Head-to-head RMSE comparison of the LSTM and S4 forecasters across all seven CHO mAb states (Table 1). Lower bars indicate lower test error.

scope of this chapter; the reader is referred to [21] for the foundational architecture.

6 Hybrid Models and Outlook

6.1 Mechanistic Priors and Learned Corrections

The sections above occupy two ends of a modeling spectrum. Kinetic ODE models (Section 2) encode biochemical mechanism directly: rate laws, stoichiometry, and conservation constraints keep the solution space physically consistent and allow extrapolation to conditions not observed during parameter estimation. Deep sequence models (Section 5) impose no such constraints; they infer temporal structure from data alone and can track complex dynamics that resist mechanistic closure, at the cost of requiring substantially more training data and providing less reliable predictions outside the training envelope.

Hybrid, or *gray-box*, models occupy the middle ground. The simplest and pedagogically cleanest construction is *residual learning*: a mechanistic model acts as a structured prior over the trajectory, and a data-driven component learns the *discrepancy* between that prior and the true data. Let $\hat{\mathbf{u}}^{\text{mech}}(t)$ denote the trajectory produced by the mechanistic model and $\mathbf{u}^*(t)$ the true (measured or simulated) trajectory. Define the residual

$$\boldsymbol{\delta}(t) = \mathbf{u}^*(t) - \hat{\mathbf{u}}^{\text{mech}}(t). \tag{45}$$

A machine-learning model f_θ is trained to map the history of $\boldsymbol{\delta}$ to its future values, and the hybrid forecast is

$$\hat{\mathbf{u}}^{\text{hybrid}}(t) = \hat{\mathbf{u}}^{\text{mech}}(t) + f_\theta(\boldsymbol{\delta}(t-L:t)), \tag{46}$$

where L is the lookback window. Several properties follow immediately. First, if the mechanistic model is exact, $\boldsymbol{\delta} \equiv 0$ and the ML component need learn nothing; the residual is zero everywhere and the hybrid reduces to the mechanistic prediction. Second, if the mechanistic model captures the large-scale qualitative shape of the trajectory but is biased (for instance because one or more parameters are mis-estimated), the ML component needs only to correct a smooth, low-amplitude signal rather than learn the full dynamics from scratch. This is the *data efficiency* argument for gray-box models: the residual is a simpler learning target than the raw state trajectory, so reliable correction requires far fewer training examples than pure data-driven forecasting would demand. Third, the mechanistic backbone provides physically consistent extrapolation beyond the training envelope; the ML correction is expected to shrink to zero as the mechanistic model becomes more accurate.

This construction is closely related to ideas in scientific machine learning [22, 23]: physics-informed neural networks embed differential equation residuals as penalty terms in the loss, while universal differential equations replace unknown rate terms with neural networks trained on trajectory data. The residual-learning framing used here is particularly natural for bioprocess engineering, where a kinetic model is almost always available but its parameters are imperfectly identified from the limited experimental data typical of process development campaigns.

6.2 Worked Example: CHO mAb Bioreactor Hybrid

To demonstrate the approach, we revisit the CHO fed-batch mAb production system of Section 2. The experimental design deliberately introduces the conditions under which a hybrid model is most relevant: the mechanistic prior is imperfect, and we assess how well the ML correction recovers the true trajectory.

Setup. The ground-truth data are the 240 h synthetic trajectories (sampled at roughly hourly resolution, with additional save points at feed-switching events, yielding 251 time points in total) generated by the full CHO kinetic model (code: `code/kinetics/cho_model.jl`, data: `code/data/cho_trajectories.cs`
To construct an imperfect mechanistic prior, two parameters are deliberately mis-specified relative to their ground-truth values: the maximum specific growth rate $\mu_{\max}$ and the biomass yield on glucose $Y_{X/\mathrm{Glc}}$ are each reduced by 25%. This is a plausible representation of an identification error — growth kinetics and yield coefficients are among the parameters most difficult to estimate reliably from batch data — and it introduces a systematic negative bias in the predicted cell density and mAb titre.

Three forecasters are trained and evaluated on the same held-out test windows (tail 20% of the 216 sliding windows derived from the 251-point trajectory with lookback $L = 24$ and horizon $H = 12$, giving 43 test windows):

1. **Pure mechanistic**: the imperfect mechanistic model is propagated forward once from $t = 0$ using the true initial conditions, and each test window's horizon forecast is read from this single biased trajectory; no data-driven correction is applied. As a result, the pure-mechanistic baseline accumulates parameter-induced model drift over the horizon, which the learned residual in the hybrid is able to remove.

2. **Pure data-driven**: an LSTM (hidden size 32, Adam at 5×10^{-3}, 300 epochs) is trained directly on the normalized true-data windows.

3. **Hybrid**: an identical LSTM is trained on normalized residual windows $\boldsymbol{\delta}$; the hybrid forecast is the imperfect mechanistic prediction plus the learned residual correction (Equation 46).

Results. Table 2 reports the per-state RMSE in physical units for all three forecasters.

Table 2: Per-state RMSE for the three forecasters on the held-out CHO mAb test windows. The mechanistic model uses parameters mis-specified by 25% in $\mu_{\max}$ and $Y_{X/\mathrm{Glc}}$. Lower is better.

State	Units	Pure Mechanistic	Pure ML (LSTM)	Hybrid
V	L	0.0090	0.0088	0.0087
X	gDW/L	0.55	0.021	0.048
[Glc]	mM	4.77	3.17	3.92
[Gln]	mM	4.13	1.05	1.17
[Lac]	mM	5.48	2.75	0.579
[Amm]	mM	2.97	0.495	0.112
[mAb]	mg/L	469	168	41.6

The hybrid corrects the mechanistic bias on every state: its RMSE is lower than that of the pure-mechanistic model in all seven dimensions, and the mean RMSE across states drops from 69.5 (pure mechanistic) to 6.78 (hybrid). For the most commercially important output, the mAb titre, the hybrid achieves 41.6 mg/L RMSE compared with 469 mg/L for the imperfect ODE — an eleven-fold reduction.

The pure-ML LSTM also corrects the mechanistic bias, as expected, and for three states (X, [Glc], [Gln]) its RMSE is somewhat lower than the hybrid. For the other four states — volume, lactate, ammonia, and most critically mAb titre — the hybrid is superior. The pattern is interpretable: states whose residual is smooth and monotonic (Lac, Amm, mAb, which reflect the cumulative bias in growth and yield) are exactly those for which the ML correction is most effective. The pure-ML model must learn the full dynamic trajectory from its training windows; the residual LSTM need only correct a slowly varying offset, which it accomplishes with the same network capacity and training budget.

Figure 8 shows the three forecasts overlaid on the true mAb trajectory for the first held-out test window, together with a per-state RMSE bar chart comparing all three approaches.

Interpretation. The key pedagogical point is not that the hybrid wins on every metric — it does not — but that it provides a *structurally different failure mode* from pure-data and pure-mechanistic forecasters. A pure-mechanistic model fails because parameter identification is imperfect; a pure-data model fails because its training distribution does not cover all operating regions; a well-designed hybrid degrades gracefully when either component fails. In this example the mechanistic prior anchors the forecast in physically plausible territory even on the held-out horizon, and the residual LSTM provides a parsimonious correction that the mechanistic model alone cannot supply.

6.3 Outlook

Identifiability and data scarcity. Bioprocess development operates under a data budget that is severe by machine learning standards. A typical fed-batch campaign yields tens of time-course experiments, each sampled at a handful of off-line assay points. Reliably estimating even a moderate kinetic model — on the order of ten to twenty parameters — from such data requires careful experimental design and structural identifiability analysis. Parameters that are not identifiable from the available measurements cannot be reliably estimated regardless of the optimization algorithm used. Hybrid models partially mitigate this constraint by reducing the number of parameters that must be estimated from data: the mechanistic backbone carries the physically constrained

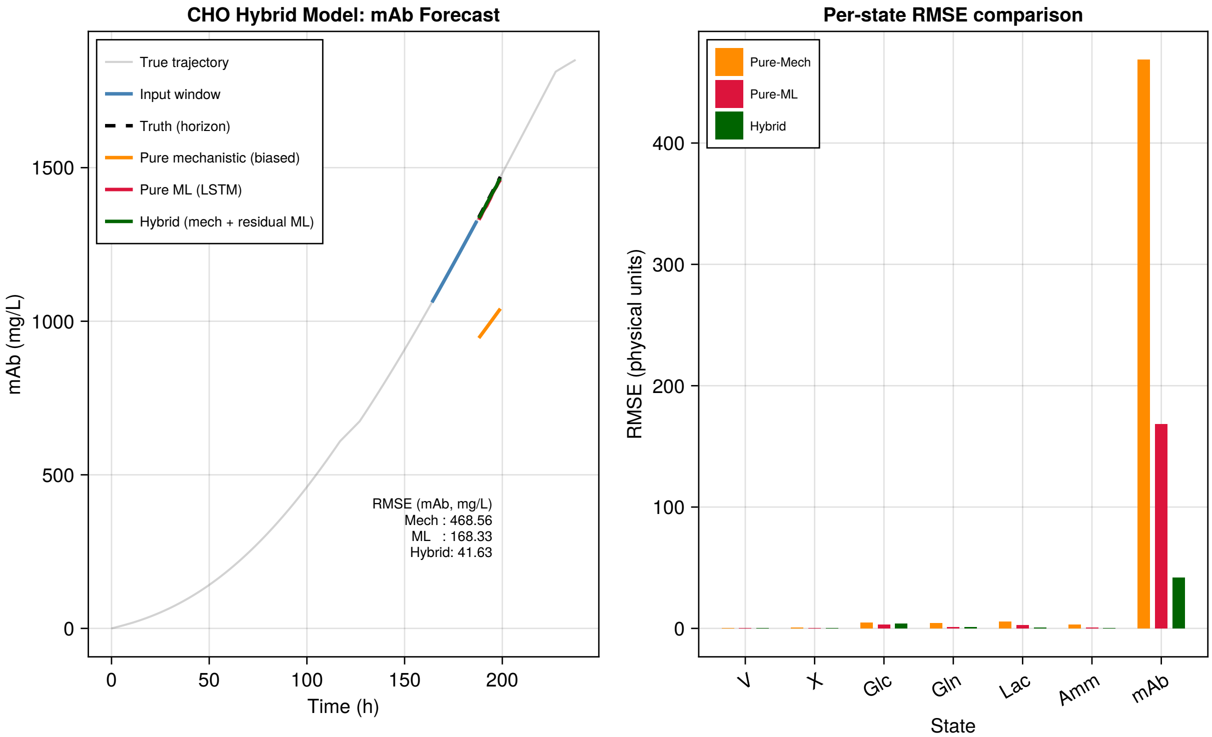

Figure 8: Three-way comparison on the CHO mAb fed-batch system. Left: true mAb trajectory (grey), 24-step input window (blue), 12-step ground-truth horizon (dashed black), pure-mechanistic forecast (orange), pure-ML LSTM forecast (red), and hybrid forecast (green) for the first held-out test window. The mechanistic model uses $\mu_{\max}$ and $Y_{X/\mathrm{Glc}}$ reduced by 25%. Right: per-state RMSE for all seven states, confirming the hybrid's bias correction across the board.

relationships, and the data-driven component need only capture the remaining discrepancy. This partitioning is effective precisely when the mechanistic structure is largely correct and the bias is compact, as in the example above.

Foundation and sequence models. The deep sequence models of Section 5 are trained from scratch on single-process data. An emerging alternative is to pre-train large sequence models on diverse bioprocess databases and fine-tune them on small process-specific datasets. This *foundation model* strategy, well established in natural language processing [21], is beginning to appear in biological and chemical applications. For bioprocess engineering the analogue would be a model pre-trained on a portfolio of fed-batch trajectories spanning multiple cell lines, media formulations, and feeding strategies, then fine-tuned with a small number of runs for a specific product. Such a model could serve as a data-efficient surrogate for process optimization and scale-up risk assessment without requiring a full mechanistic closure.

Protein and genome sequence models [24] are also reshaping the upstream of bioprocess design. If cell-line productivity can be predicted from sequence features, the classical separation between strain engineering and process engineering begins to dissolve. Integrating sequence-level predictions of productivity, growth rate, and metabolic phenotype with the kind of fed-batch process models developed in Sections 2 through 5 is an open and active research direction.

Digital twins for bioprocesses. A *digital twin* is a computational model that runs in correspondence with a physical process, updating its state as new measurements arrive and providing predictions used for real-time decision support. The mechanistic ODE model of Section 2 is a minimal digital twin: it can be re-initialized or re-parameterized from on-line measurements and used to project forward in time for feeding-strategy optimization or end-of-batch titer prediction. Hybrid architectures extend this concept by allowing the twin to correct its own mechanistic bias using residuals computed from real-time sensor data, approaching the behavior of a continuously updating soft sensor. In conjunction with the gene-expression modules of Section 4, a full-scale digital twin might couple the metabolic state of the culture to the transcriptional regulatory network, enabling prediction of how transient stress responses feed back into productivity.

Synthesis. This chapter has traced an arc from first-principles kinetics through constraint-based metabolic models, regulatory network motifs, deep sequence forecasters, and gray-box hybrids. The arc is not a progression in which each paradigm supersedes the last. Mechanistic models provide the physical constraints and extrapolation capability that data-driven models lack; data-driven models provide the flexibility and scalability that mechanistic models cannot match; hybrid architectures are a principled attempt to draw on the strengths of both. The worked examples fall into two groups. The CHO fed-batch mAb bioreactor is the shared running example of Sections 2, 5, and 6: the kinetic ODE model supplies ground truth, the deep time-series forecasters learn to predict that trajectory, and the residual-learning hybrid corrects a deliberately mis-specified mechanistic prior — three complementary lenses trained on the same biological system, which together demonstrate that no single lens is sufficient. The FBA urea-cycle case study (Section 3) and the network-motif and cybernetic examples (Section 4) are complementary illustrations of distinct modeling paradigms — constraint-based flux analysis and regulatory-circuit design principles — applied to different biological contexts, broadening the conceptual scope of the chapter beyond what any single system could illustrate.

A Derivations

A.1 Quasi-Steady-State Derivation of the Michaelis–Menten Rate Law

Consider the single-enzyme mechanism

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P, \tag{47}$$

where E, S, ES, and P denote free enzyme, substrate, enzyme–substrate complex, and product, respectively. Writing mass-action balances for each species:

$$\frac{d[E]}{dt} = -k_1[E][S] + (k_{-1} + k_2)[ES], \tag{48}$$

$$\frac{d[ES]}{dt} = k_1[E][S] - (k_{-1} + k_2)[ES], \tag{49}$$

$$\frac{d[S]}{dt} = -k_1[E][S] + k_{-1}[ES], \tag{50}$$

$$\frac{d[P]}{dt} = k_2[ES]. \tag{51}$$

The total enzyme concentration $[E]_T = [E] + [ES]$ is conserved, so $[E] = [E]_T - [ES]$.

The quasi-steady-state assumption (QSSA). When the enzyme is present at a concentration much lower than the substrate ($[E]_T \ll [S]_0$), the complex ES reaches a pseudo-steady state on a fast time scale relative to the depletion of S. Setting $d[ES]/dt \approx 0$ in Eq. (49) gives:

$$k_1([E]_T - [ES])[S] = (k_{-1} + k_2)[ES]. \tag{52}$$

Solving for $[ES]$:

$$[ES]\big(k_1[S] + k_{-1} + k_2\big) = k_1[E]_T[S], \tag{53}$$

$$[ES] = \frac{k_1[E]_T[S]}{k_1[S] + k_{-1} + k_2} = \frac{[E]_T[S]}{[S] + \dfrac{k_{-1} + k_2}{k_1}}. \tag{54}$$

Defining the Michaelis constant

$$K_M \equiv \frac{k_{-1} + k_2}{k_1}, \tag{55}$$

Eq. (54) becomes $[ES] = [E]_T[S]/(K_M + [S])$.

The rate law. The product formation rate (velocity) $v = d[P]/dt = k_2[ES]$ is therefore:

$$v = \frac{k_2[E]_T[S]}{K_M + [S]} = \frac{V_{\max}[S]}{K_M + [S]}, \tag{56}$$

where $V_{\max} \equiv k_2[E]_T$. This is the Michaelis–Menten equation [1].

Limiting behavior.

- $[S] \ll K_M$: $v \approx (V_{\max}/K_M)[S]$, a first-order dependence on substrate with apparent first-order rate constant $k_{\text{cat}}/K_M = k_2/K_M$.
- $[S] \gg K_M$: $v \approx V_{\max}$, zero-order in substrate; all enzyme active sites are saturated.
- $[S] = K_M$: $v = V_{\max}/2$, the defining operational meaning of K_M.

Validity of the QSSA. The QSSA is formally valid when $[E]_T \ll K_M + [S]_0$, where $[S]_0$ is the initial substrate concentration (note this is a tighter condition than the classical $[E]_T \ll [S]_0$, which neglects the contribution of K_M). Under these conditions the time scale for ES to equilibrate ($\tau_{ES} \sim 1/(k_1[S] + k_{-1} + k_2)$) is much shorter than the substrate depletion time ($\tau_S \sim [S]_0/V_{\max}$), justifying the pseudo-steady-state approximation.

A.2 Derivation of the Steady-State Flux-Balance Constraint $\mathbf{S}\,\mathbf{v} = \mathbf{0}$

Consider a well-mixed intracellular compartment of volume V containing m metabolite species and r enzymatic and transport reactions. Let x_i (mol) denote the moles of species i inside the compartment. An open-species mole balance for species i accounts for chemical conversion, transport across the compartment boundary, and dilution by cell growth:

$$\frac{dx_i}{dt} = \sum_{j=1}^{r} S_{ij}\, v_j\, V - \mu\, x_i, \qquad i = 1, \ldots, m, \tag{57}$$

where S_{ij} is the stoichiometric coefficient of species i in reaction j (negative for reactants, positive for products), v_j is the volumetric reaction rate of reaction j (mol per unit volume per unit time), and μ is the specific growth rate. The term $S_{ij}\, v_j\, V$ is the molar rate of production of species i by reaction j; the term $\mu\, x_i$ accounts for the dilution of intracellular species as the cell grows and the compartment volume increases.

Expressing concentrations $c_i = x_i/V$ and assuming constant volume (constant cell size during balanced growth), Eq. (57) becomes

$$\frac{dc_i}{dt} = \sum_{j=1}^{r} S_{ij}\, v_j - \mu\, c_i, \qquad i = 1, \ldots, m, \tag{58}$$

or in matrix form

$$\frac{d\mathbf{c}}{dt} = \mathbf{S}\,\mathbf{v} - \mu\,\mathbf{c}. \tag{59}$$

Separation of time scales. Metabolic fluxes equilibrate on time scales of seconds to minutes; the cell-growth time scale $\tau_\mu = \mu^{-1}$ is typically hours. Consequently, $|\mathbf{S}\,\mathbf{v}| \gg |\mu\,\mathbf{c}|$ for most intracellular metabolites, and the dilution term is negligible relative to the metabolic production and consumption terms. On the fast metabolic time scale $\mathbf{c}$ therefore relaxes to a pseudo-steady state satisfying

$$\frac{d\mathbf{c}}{dt} \approx \mathbf{0}. \tag{60}$$

The steady-state constraint. Substituting Eq. (60) into Eq. (59) and dropping the negligible dilution term gives

$$\boxed{\mathbf{S}\,\mathbf{v} = \mathbf{0}.} \tag{61}$$

This is the fundamental steady-state mass-balance constraint of flux balance analysis: the stoichiometric matrix $\mathbf{S}$ maps the flux vector $\mathbf{v}$ to zero, so $\mathbf{v}$ must lie in the (right) null space $\mathcal{N}(\mathbf{S})$.

For a network with $r > m$ (more reactions than metabolites) the null space has positive dimension $r - \text{rank}(\mathbf{S}) > 0$, meaning that infinitely many flux vectors satisfy Eq. (61). Flux balance analysis resolves this underdeterminacy by selecting the element of $\mathcal{N}(\mathbf{S})$ that maximizes a biologically motivated linear objective, subject to inequality bounds on individual fluxes, as described in Section 3.

A.3 Derivation of the Hill Input Function

Consider a promoter with n identical, independent binding sites for a transcription factor (TF) monomer X. In the cooperative limit each of the n sites must be occupied simultaneously for the promoter to be active (all-or-nothing binding). Let p denote the probability that any single site is occupied. At equilibrium $p = x/(K_d + x)$, where x is the free TF concentration and K_d is the microscopic dissociation constant. The probability that all n sites are simultaneously occupied is

$$P_{\text{on}} = p^n = \frac{x^n}{(K_d + x)^n}. \tag{62}$$

In the all-or-nothing (two-state) approximation, intermediate partially-bound states are neglected and the bound fraction takes the Hill form $P_{\text{on}} \approx x^n/(K^n + x^n)$, with the half-saturation constant $K \approx K_d$. This is the activating Hill function with n as the cooperativity (Hill) coefficient and K as the half-maximal activation concentration [10]. The same derivation via a thermodynamic (Boltzmann) partition function over all 2^n binding states yields

$$f_{\text{act}}(x) = \frac{(x/K)^n}{1 + (x/K)^n} = \frac{x^n}{K^n + x^n}, \tag{63}$$

confirming the form in Eq. (16). The repressing Hill function $f_{\mathrm{rep}}(x) = 1 - f_{\mathrm{act}}(x) = K^n/(K^n + x^n)$ describes the complementary case in which TF binding occludes the promoter. When $n = 1$ the functions reduce to simple hyperbolic (Michaelis-type) saturation; for $n > 1$ the response becomes progressively more switch-like, approaching a step function at $x = K$ as $n \to \infty$.

A.4 C1-FFL Sign-Sensitive Delay and I1-FFL Pulse Analysis

C1-FFL sign-sensitive delay. In the coherent type-1 FFL with AND gate at Z, the output is governed by Eqs. (20)–(21) in Section 4.2. When X steps on at $t = 0$, the direct arm immediately drives the AND-gate factor $f_{\mathrm{act}}(X;\, K_{XZ},\, n)$ toward 1, but the second AND-gate factor $f_{\mathrm{act}}(Y;\, K_{YZ},\, n)$ remains near zero until Y rises above K_{YZ}. Since Y relaxes exponentially toward its steady state $Y^* = \beta_Y/\alpha_Y$ with time constant $\tau_Y = 1/\alpha_Y$, the time at which Y crosses K_{YZ} is approximately

$$t_{\mathrm{delay}} \approx \frac{1}{\alpha_Y} \ln\left(\frac{Y^*}{Y^* - K_{YZ}}\right), \tag{64}$$

giving the sign-sensitive delay for sustained ON inputs. For a transient pulse of duration $\Delta t \ll 1/\alpha_Y$, Y barely rises from zero before X turns off again; the AND gate never opens and Z is filtered. The delay appears only for the ON transition (sign-sensitive), not the OFF transition, in which Z falls immediately once the direct arm is removed. This asymmetry allows the C1-FFL to discriminate between brief noise and sustained environmental signals [14].

I1-FFL pulse analysis. In the incoherent type-1 FFL governed by Eqs. (22)–(23), let X step on at $t = 0$. Immediately after the step the repressor Y is still zero, so $f_{\mathrm{rep}}(0;\, K_{YZ},\, n) = 1$ and Z is driven at its maximal rate $\beta_Z\, f_{\mathrm{act}}(X;\, K_{XZ},\, n)$. On the time scale $1/\alpha_Y$, Y accumulates to $Y^* = \beta_Y\, f_{\mathrm{act}}(X;\, K_{XY},\, n)/\alpha_Y$ and suppresses Z. The new Z steady state is

$$Z^* = \frac{\beta_Z}{\alpha_Z}\, f_{\mathrm{act}}(X;\, K_{XZ},\, n)\, f_{\mathrm{rep}}(Y^*;\, K_{YZ},\, n) < Z_{\mathrm{peak}}, \tag{65}$$

so Z first rises to a transient peak before falling back to Z^*. The peak height is set by the early dynamics when $Y \approx 0$, and Z^* is reduced relative to the peak by the factor $f_{\mathrm{rep}}(Y^*;\, K_{YZ},\, n)$. The fold-change detection property [15] follows from the observation that if X is multiplied by a constant factor λ (fold-change), both Y^* and the initial rate of Z production (dZ/dt at $t = 0^+$) scale proportionally, leaving the ratio Z_{peak}/Z^* invariant—the circuit reports the *ratio* of signal levels, not their absolute magnitudes.

A.5 Cybernetic Allocation Objective

The cybernetic control variables u_i and v_i can be derived from a growth-rate maximization principle [17]. At each instant the cell must allocate a fixed total enzyme-synthesis capacity r_E across N pathways. Let $u_i \geq 0$ with $\sum_i u_i = 1$ denote the synthesis fractions, and let $v_i \in [0, 1]$ denote activity fractions. The instantaneous growth rate is

$$\mu = \sum_{i=1}^{N} rG_i\, v_i, \qquad rG_i = \mu_{\max,i}\, \frac{E_i}{E_{\max,i}}\, \frac{S_i}{K_i + S_i}. \tag{66}$$

Maximizing μ over $v_i \in [0, 1]$ subject to no constraint other than the bound gives $v_i = 1$ for the pathway with the largest rG_i and $v_i = 0$ for all others—an all-or-nothing LP solution. The matching-law variant smooths this discontinuous policy by solving a proportional-share problem:

the synthesis allocation u_i that maximizes the expected future return under uncertainty about which substrate will persist is proportional to the current relative return

$$u_i = \frac{rG_i}{\sum_j rG_j}, \tag{67}$$

which is the matching law of behavioral ecology applied to enzyme synthesis. Enzyme activity is modulated by the normalized relative advantage,

$$v_i = \frac{rG_i}{\max_j rG_j}, \tag{68}$$

so the best pathway operates at full activity ($v = 1$) and all others are partially suppressed. These two rules together constitute the cybernetic model as implemented in `cybernetic.jl` and governed by Eqs. (31)–(33) of Section 4.3.

A.6 Backpropagation Through Time and the Vanishing-Gradient Problem

Consider an Elman RNN with scalar hidden state for clarity. At step t the state is $h_t = \phi(w_{hh}\, h_{t-1} + w_{uh}\, u_t + b)$ and the output is $\hat{y}_t = w_{yh}\, h_t$. The scalar loss over a sequence of length T is $\mathcal{L} = \sum_{t=1}^{T} \ell(\hat{y}_t, y_t)$.

Unrolling. Backpropagation through time unrolls the recurrence into a computation graph with T layers, each parameterized by the same weights. By the chain rule, the gradient of the loss at the final step with respect to the hidden state at step t is

$$\frac{\partial \mathcal{L}}{\partial h_t} = \frac{\partial \mathcal{L}}{\partial h_T} \prod_{k=t+1}^{T} \frac{\partial h_k}{\partial h_{k-1}}, \tag{69}$$

where

$$\frac{\partial h_k}{\partial h_{k-1}} = \phi'(w_{hh}\, h_{k-1} + w_{uh}\, u_k + b) \cdot w_{hh}. \tag{70}$$

Vanishing and exploding gradients. The product in Eq. (69) contains $T - t$ factors of the form $\phi'(\cdot)\, w_{hh}$. For the tanh activation, $|\phi'(\cdot)| \leq 1$, so if $|w_{hh}| < 1$ the product decays geometrically as $O\big((|w_{hh}|)^{T-t}\big) \to 0$ as $T - t \to \infty$. Gradients flowing to early time steps effectively vanish, preventing the model from learning dependencies spanning many steps. Conversely, if $|w_{hh}| > 1$ the product grows without bound, causing numerically unstable gradient explosions. In the multi-dimensional case the relevant quantity is the spectral radius $\rho(W_{hh}\, \mathrm{diag}[\phi'(\mathbf{h}_k)])$: if this exceeds one the gradients explode; if it is less than one for all k the gradients vanish. Gradient clipping addresses explosions; the LSTM architecture (Section 5.2) addresses vanishing gradients structurally.

A.7 LSTM Gate Equations and Gradient Flow

The LSTM update equations for step t are reproduced from Eqs. (35)–(40) for convenience. Let $\mathbf{z}_t = [\mathbf{h}_{t-1}; \mathbf{u}_t]$ denote the concatenated input. Then

$$\mathbf{i}_t = \sigma(W_i\,\mathbf{z}_t + \mathbf{b}_i), \tag{71}$$

$$\mathbf{f}_t = \sigma(W_f\,\mathbf{z}_t + \mathbf{b}_f), \tag{72}$$

$$\mathbf{o}_t = \sigma(W_o\,\mathbf{z}_t + \mathbf{b}_o), \tag{73}$$

$$\tilde{\mathbf{c}}_t = \tanh(W_c\,\mathbf{z}_t + \mathbf{b}_c), \tag{74}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \tag{75}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t). \tag{76}$$

Gradient flow through the cell state. The key insight is that the cell-state update Eq. (75) involves an *additive* term: the new cell state is the old cell state scaled by the forget gate plus a new contribution. The partial derivative of $\mathbf{c}_t$ with respect to $\mathbf{c}_{t-1}$ is

$$\frac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} = \mathrm{diag}(\mathbf{f}_t), \tag{77}$$

a diagonal matrix of forget-gate values. The corresponding BPTT product for the cell state is $\prod_{k=t+1}^{T} \mathrm{diag}(\mathbf{f}_k)$. Because the forget gates are *learned* sigmoid activations, the network can set $\mathbf{f}_k \approx \mathbf{1}$ on channels where long-range memory is required, keeping the product close to the identity and preventing gradient decay. This is the structural mechanism by which the LSTM avoids the vanishing gradient: information flows through the cell state along a highway of forget-gate-modulated identity connections [19].

A.8 S4 Bilinear Discretization and the HiPPO-LegS Construction

Bilinear (Tustin) discretization. Starting from the continuous-time linear system $\dot{\mathbf{x}}(t) = A\,\mathbf{x}(t) + B\,\mathbf{u}(t)$, integrate over one step $[t_{k-1}, t_k]$ of length Δt using the trapezoidal approximation $\int_{t_{k-1}}^{t_k} A\mathbf{x}\,d\tau \approx \frac{\Delta t}{2}\big(A\mathbf{x}_k + A\mathbf{x}_{k-1}\big)$, which gives

$$\mathbf{x}_k - \mathbf{x}_{k-1} = \tfrac{\Delta t}{2}\,A\,\mathbf{x}_k + \tfrac{\Delta t}{2}\,A\,\mathbf{x}_{k-1} + \Delta t\,B\,\mathbf{u}_k. \tag{78}$$

Collecting $\mathbf{x}_k$ terms on the left:

$$\big(I - \tfrac{\Delta t}{2}A\big)\,\mathbf{x}_k = \big(I + \tfrac{\Delta t}{2}A\big)\,\mathbf{x}_{k-1} + \Delta t\,B\,\mathbf{u}_k. \tag{79}$$

Provided $(I - \frac{\Delta t}{2}A)$ is invertible, which holds for all finite Δt when the eigenvalues of A are finite, the discrete recurrence is

$$\mathbf{x}_k = \bar{A}\,\mathbf{x}_{k-1} + \bar{B}\,\mathbf{u}_k, \tag{80}$$

with

$$\bar{A} = \big(I - \tfrac{\Delta t}{2}A\big)^{-1}\big(I + \tfrac{\Delta t}{2}A\big), \qquad \bar{B} = \big(I - \tfrac{\Delta t}{2}A\big)^{-1}\Delta t\,B. \tag{81}$$

This is the bilinear (Tustin) transformation, Eq. (42) of Section 5.2.3. The bilinear rule is preferred over the simpler forward Euler ($\bar{A} = I + \Delta t A$) because it preserves stability: if A has all eigenvalues in the open left half-plane, the bilinear map sends them into the open unit disk, so the discrete system is also stable.

HiPPO-LegS construction. The HiPPO (High-order Polynomial Projection Operators) framework of Gu et al. [20] seeks a pair (A, b) such that the continuous-time recurrence $\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + b\,u(t)$ maintains an optimal polynomial approximation of the input history $u(\tau)$, $\tau \leq t$. Under the Legendre scaled (LegS) measure — in which the approximation is measured in $L^2([0, t])$ with the uniform weight, and the approximation order h is held fixed as t grows — the optimal matrices are

$$A_{nk} = -\begin{cases} \sqrt{(2n+1)(2k+1)} & n > k, \\ n+1 & n = k, \\ 0 & n < k, \end{cases} \qquad b_n = \sqrt{2n+1}, \quad n, k = 1, \ldots, h. \tag{82}$$

The matrix A is lower-triangular with negative diagonal, ensuring that all eigenvalues are negative real numbers and that the continuous system is asymptotically stable. The latent state $\mathbf{x}(t)$ at any time t holds the optimal Legendre coefficients for the input up to t; reading out a linear combination $C\mathbf{x}(t)$ therefore reconstructs a polynomial approximation of any desired functional of the input history. This is the theoretical justification for training only C (and D) while keeping (A, B) frozen: the structured state already encodes all the long-range information present in the input window, and the readout selects the relevant combination for the forecasting task [20].

In the implementation (`code/deeplearning/s4.jl`), one independent h-dimensional LegS filter is applied per input channel in a block-diagonal arrangement so that the channels do not interact through the state matrix. The block-diagonal $A \in \mathbb{R}^{(hd_{\text{in}})\times(hd_{\text{in}})}$ and $B \in \mathbb{R}^{(hd_{\text{in}})\times d_{\text{in}}}$ are constructed by `build_legS_matrices_mimo`, discretized by `discretize_bilinear`, and then frozen. Only the readout $C \in \mathbb{R}^{(N_{\text{states}}\times H_{\text{horizon}})\times N_{\text{lat}}}$ and feedforward D are updated during training.

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